

Improving the Early Childhood Environment: Direct and Distributional Effects on Human Capital for Multiple Generations^{*}

Tania Barham, Brachel Champion, Gisella Kagy, and Jena Hamadani[†]

May 19, 2025

Abstract

We demonstrate that improving the early childhood health environment has long-term and intergenerational effects on human capital. We take advantage of a basic public health program in Matlab Bangladesh, which included childhood vaccinations such as measles, that was quasi-randomly placed into treatment and comparison areas. We show adults who were eligible for the program as children exhibit increased height and achieved higher levels of educational attainment. Findings are largest for individuals with the lowest pre-program health endowment, revealing a 50% reduction in short-stature and 0.2 SD increase in cognition, reducing disadvantage across generations. Intergenerational analysis reveals that daughters of eligible, compared to ineligible, mothers were taller, less likely to be stunted, and had higher cognition. These effects are correlated with mother's agriculture income generation activities. While the direct impacts on eligible children are substantial, the intergenerational effects demonstrate how early health investments continue to yield returns across generations. These findings suggest that cost-benefit analyses of early childhood health programs that ignore long-term or intergenerational effects underestimate their full returns.

JEL Classification: I14, I15, J13, O12

Keywords: human capital, intergenerational, vaccination, Bangladesh

^{*}We thank Sajeda Amin, Francisca Antman, Jason Boardman, Ryan Brown, Brian Cadena, Chloe East, Andrew Foster, Chris Jochem, Randall Kuhn, Richard Mansfield, Terra McKinnish, Jane Menken, John Maluccio, Hani Mansour, Noboko Mizoguchi, Mushfiq Mobarak, Colleen Reid, Elisabeth Root, Paul Schultz, Duncan Thomas, Patrick Turner, and Andrea Velásquez. The data collection for this project was generously funded by the National Institutes of Health, Population Research Bureau, the International Initiative for Impact Evaluation, and the CU Population Center. We are deeply grateful to our many staff partners at icddr,b for their partnership and data access. The Matlab Health and Socioeconomic Survey-wave 2 (MHSS2), will be made publicly available at ICPSR alongside the 1996 Matlab Health and Socioeconomic Survey (MHSS1), and will include some of the icddr,b data that the authors link to in this paper. The views expressed in this article are those of the authors and do not necessarily reflect the official policy or position of the United States Air Force Academy, the Air Force, the Department of Defense, or the U.S. Government.

[†]Barham: University of Colorado Boulder, Economics Building Rm 212, 256 UCB, Boulder, CO 80309 (tania.barham@colorado.edu); Champion: United States Air Force Academy, 2354 Fairchild Drive, Suite 6K-110, Colorado Springs, CO 80840 (brachel.champion@afacademy.af.edu); Kagy: University of Wisconsin - Madison, 1300 Linden Dr, Madison, WI 53706 (gkagy@wisc.edu); Hamadani: icddr,b (jena@icddr.org).

1 Introduction

A substantial body of evidence documents strong links between the human capital of parents and their children.¹ Economic models of human capital formation and transmission suggest that these intergenerational correlations arise from a complex interplay of biological and genetic factors, environmental conditions, and parental investments in children.² These models also predict that positive investments during sensitive periods of child development—such as early childhood health—can have long-lasting effects on human capital, improving economic opportunities in adulthood and for subsequent generations. While many public health programs focus on improving early childhood health—and global investment in maternal and child health has increased in recent decades—future funding remains uncertain in some contexts. This uncertainty underscores the importance of quantifying the long-term and intergenerational impacts of these investments to provide a more comprehensive understanding of their returns and assess whether they can help break the cycle of disadvantage across generations.

In this paper, we bring new panel data that spans three generations to determine the long-term and intergenerational effects of public health interventions and if these effects vary by one's pre-program health endowment. To identify the effects, we take advantage of the quasi-randomly placed Maternal and Child Health and Family Planning (MCH-FP) Program. The MCH-FP program was rolled out over time in the Matlab district of Bangladesh in the late 1970s and 80s in a treatment area, leaving a comparison area for evaluation purposes. The services provided are key inputs of a basic public health package, including the provision of vaccines against debilitating early childhood diseases such as measles, polio, pertussis, maternal tetanus toxoid vaccination, and modern contraception (Bhatia et al., 1980). Together, these interventions represent a substantial improvement in the early childhood health environment.

We investigate the impacts of the MCH-FP program on the human capital of two generations. Long-term effects are evaluated through the first-generation cohort, which comprises individuals born between 1982 and 1988—a period when both modern contraception and intensive early child health interventions were accessible in the treatment area. First-generation long-term human capital outcomes are assessed when individuals are approximately 24 to 30 years old. We further investigate the distributional effects based on pre-program health endowments, proxied by the height of the mothers of the first generation,

1. For examples on income and education see Black and Devereux (2011), Hertz et al. (2008), Richter and Robling (2013), for health Bhalotra and Rawlings (2013) and Venkataramani (2011), and for cognition Grönqvist, Öckert and Vlachos (2017).

2. For theoretical models of early-life skill formation in economics, see Attanasio (2015), Cunha and Heckman (2007), and Heckman (2007*a*); for biological models of health and disease, see (Gluckman, Buklijas and Hanson, 2016).

whom we refer to as generation zero. Intergenerational effects are measured using the second-generation cohort, which includes the firstborn children of females in the first-generation cohort. Their human capital outcomes are measured at 0–14 years of age. We present a conceptual framework, drawing from the dynamic human capital production function model of Heckman (2007a) and Attanasio, Meghir and Nix (2020), to illustrate how these three generations may be interconnected and how the MCH-FP program can influence human capital within and across generations.

The scope and rigor of the study are enhanced by the rich data that links individuals over time across several data sources, connecting three generations. Outcomes are drawn from the 2012 Matlab Health and Socioeconomic Survey 2 (MHSS2) and provide information on multiple measures of human capital: height, short stature/stunting, cognition, and grades attained. MHSS2 has low levels of attrition—less than 10 percent—despite data being collected 35 years after the program start. To examine the robustness of the research design, these data are linked to several data sets on the study area including MHSS1 from 1996, demographic surveillance data (including births, deaths, and migration) that span the pre-program period to MHSS2, and pre-program census data of the study area.

A key feature of the MCH-FP program is that treatment and comparison areas were built into the program and designed to be economically and socially similar (Figure 1). Villages were purposefully not randomized into treatment, rather villages were placed into six blocks created by the program to facilitate program delivery and limit potential substantial spillovers from vaccination and information sharing. The block design was also essential for the efficacy of the vaccines since some of the diseases (e.g. measles) are highly contagious and require a high percent of the population to be vaccinated to eliminate the disease. We take advantage of the well-designed treatment and comparison areas to estimate intent-to-treat (ITT) single-difference models for both the long-run and intergenerational effects.³ Rich data availability, including a pre-program census, allows us to provide multiple pieces of evidence to demonstrate the comparison area is a good counterfactual. First, we show the means of height, education and cognition for the parents of our first generation cohort are similar between the treatment and comparison areas. Second, we show the pre-program cohort trends between the treatment and comparison areas are similar for education. Third, we show there are few pre-program differences in means of other variables that

3. There is no pre-analysis plan for this research as the data collection for this paper was conceived prior to their common use in economics. Matlab Health and Socio-Economic Survey 2 (MHSS2), was designed and collected by some of the authors in collaboration with a multi-disciplinary group, to study the long-term effects of the MCH-FP program. Descriptions of the planned research design and analysis are in the grant applications for the data collection.

could affect human capital (e.g. most individual and household characteristics). Lastly, we refer to existing literature showing that for the entire Matlab area migration trends, fertility and mortality were similar between the treatment and comparison areas prior to the program. We further show there is no evidence of spatially correlated errors between treatment and comparison areas, and results are robust to multiple hypothesis testing, randomization-based inference that takes into account the placement of villages into a contiguous treatment area, and using a subset of the comparison area villages. Finally, we present double-difference models for both generations using less-affected cohorts to control for potential baseline differences and results are again similar.

Long-term effects on the first generation reveal human capital gains documented in late childhood (Barham, 2012; Joshi and Schultz, 2013) persist into adulthood for height and grades attained but not for cognition.⁴ Both males and females are about one centimeter taller. In addition, effects for grades attained are almost one year for men. There are no program effects on education for first generation females which is consistent with the availability of a successful nationwide girl's secondary school scholarship program operating in both the treatment and comparison areas at the time (Shamsuddin, 2015).

To understand the distributional effects of the program on the first generation, we examine treatment effects by the individual's pre-program health endowment. We use stunted mothers (generation zero) as a proxy indicator of a low endowment for the first generation. Generation zero's height is not affected by the program and is reflective of the health environment when generation zero was a child, and thus is a good proxy for the health endowment they pass on to their children. Results show program effects for the first generation are largest for those with the lowest health endowment, where potential gains from health improvements were the largest. First generation females born to these mothers had height improvements of 2.9cm and a 64% reduction in short stature. These findings indicate the MCH-FP program contributed to a reduction in human capital gaps for the first generation.

Intergenerational effects are concentrated solely among second-generation females. ITT effects indicate that female offspring of first generation treated mothers are on average taller (0.34 SD) and have higher levels of cognition (0.17 SD). These height gains are observed at the lower end of the distribution, resulting in a sizable 50 percent reduction in stunting, underscoring that those who were potentially the most disadvantaged benefited from the program.

4. Barham (2012) shows that older cohorts born before 1982—who were exposed to the program aged 15 or older—experienced no statistically significant effects.

We explore several behavioral mechanisms that could drive the sex-specific intergenerational effects. We do not find evidence that fertility or mortality selection by sex, that could arise from the MCH-FP program itself, drive the results. We further rule out results are due to differential investment in typical pre- and post-natal child health interventions, such as pre-natal care visits, skill of birth attendant, or vaccination. We further examine differences in parental characteristics that could affect their child investment behavior. ITT effects indicate that mothers of second-generation females are twice as likely to generate agricultural income as mothers of second-generation males. While this sizable difference is estimated with some imprecision, it suggests that mothers with firstborn daughters may have greater economic and nutritional resources available to invest in their daughters' human capital. This differential investment could help explain the concentration of positive effects among second-generation females.

Our study makes four key contributions. First, we document significant and persistent effects—both in the long term and across generations—of early childhood health and family planning interventions that have been routinely implemented worldwide. Rigorous causal evidence on the long-term and intergenerational impacts of such interventions is scarce, largely due to the lack of well-designed studies that follow individuals and their children over several decades and have sufficient data availability. This is evidenced by the lack of randomized controlled trials examining long-term and intergenerational impacts of typical public health interventions during childhood.

While the literature on long-term effects is expanding,^{5,6} causal intergenerational evidence remains limited.⁷ Important exceptions include studies exploiting the state-level rollout of Medicaid (East et al., 2023) and the measles vaccine (Noghanibehambari, 2024) in the U.S., which use secondary data on maternal birth outcomes and show reductions in the incidence of low birth weight. In a lower-income setting, Behrman et al. (2009) show that early childhood nutrition interventions in Guatemala improved

5. Evidence on the long-term effects of early child health improvements on adult human capital outcomes in the U.S. focuses on state expansions of large federally funded programs, such as Medicaid (Miller and Wherry, 2019), food stamps (Hoynes, Schanzenbach and Almond, 2016), and the measles vaccine (Bartleska et al., 2023), and demonstrates improved educational outcomes. In addition, Hoynes, Schanzenbach and Almond (2016) show improved adult health, including reductions in chronic disease and the incidence of metabolic syndrome. However, evidence on cognitive outcomes is limited. Further, Bailey (2013) highlights that family planning policies can also increase college completion decades later. In developing countries, Maluccio et al. (2009) and Walker et al. (2021) show that early childhood nutrition interventions lead to higher educational attainment, improved reading comprehension, and better cognitive outcomes in adulthood.

6. Barham, Kuhn and Turner (2023) use MHSS2 to estimate MCH-FP effects on labor and migration patterns and key mechanisms for the first generation, including education. Results on education are similar (see Section 2.4 for more details).

7. Early intergenerational literature primarily focuses on negative historical shocks, such as famines or disease outbreaks, which are not necessarily symmetric to targeted positive interventions. These shocks typically affected more than just the targeted child and mother, and their impacts were arguably more severe (Painter et al., 2008; Almond et al., 2010).

height- and weight-for-age for male offspring of participating mothers, although attrition was high.⁸ Related research on interventions targeting slightly older children demonstrates intergenerational improvements in educational attainment from early education programs in the U.S., such as Head Start (Barr and Gibbs, 2017) and the Perry Preschool Project (Garcia, Heckman and Ronda, 2021), as well as reductions in childhood mortality from a school-based deworming program in Kenya (Walker et al., 2023).

Second, our study advances the understanding of intergenerational effects in several important ways. We examine multiple measures of human capital beyond birth outcomes. This is essential, as many behavioral mechanisms linking generations—such as postnatal parental investments—manifest after birth and may affect different dimensions of human capital in distinct ways. Moreover, prior literature does not systematically explore differences by sex, despite growing evidence from the behavioral and biological sciences suggesting sex-specific mechanisms (Drake and Liu, 2010; Sandovici et al., 2022). Indeed, we find stronger effects among second-generation females than males. Finally, the availability of rich panel data allows us to explore mechanisms beyond what is feasible in many other settings. High-frequency secondary data enable us to rule out sample selection due to differential fertility and mortality caused by the program. Additionally, extensive individual-level survey data provide suggestive evidence that women’s labor supply may play a role in driving the intergenerational effects.

Third, this study is unique in identifying where in the health endowment distribution the program’s effects are most pronounced—offering insights into changes in human capital inequality. We show that individuals in the first generation with the lowest levels of maternal health endowment benefit the most from an improved early childhood environment. This finding suggests that such interventions can reduce development gaps in human capital outcomes in adulthood and potentially break the intergenerational transmission of disadvantage.

Finally, we address several methodological challenges commonly encountered in long-term and intergenerational research. Notably, the inclusion of comparable treatment and control areas in the program design, along with the use of panel data, is essential for estimating intent-to-treat (ITT) effects, mitigating selection into treatment, and controlling for underlying time trends over such long periods of time. Additionally, the attrition rate in our study is remarkably low. This is essential for minimizing selection bias arising from key sources of attrition—such as differential mortality and migration—that is often caused by the program itself.

8. Results are based on panel survey data that tracked approximately 60 percent of the original sample.

2 MCH-FP Program

2.1 MCH-FP Intervention

The MCH-FP program began in October 1977 in the rural Matlab district of Bangladesh and was administered by icddr,b, (formally known as International Center for Diarrhoeal Disease Research, Bangladesh). The program included integrated family planning and maternal and child health services. A key feature of the program was that interventions were free and administered in the beneficiary's home during monthly visits made by local female health workers hired and trained by the program (Bhatia et al., 1980).

Services were implemented in two main phases: October 1977 to February 1982 and March 1982 to December 1988. During phase one, interventions focused on family planning and maternal health through the provision of modern contraception, tetanus toxoid vaccinations for pregnant women (starting June 1978), and iron and folic acid tablets for women in the last trimester of pregnancy (Bhatia et al., 1980). Tetanus toxoid was expanded to all women of reproductive age in 1982. During these visits, the female health workers also provided counseling on contraceptives, nutrition, hygiene, and breastfeeding, and provided instructions on oral rehydration solution preparation. Follow-up and referral systems ensured management of side effects and aided in the continued use of contraceptives (Phillips et al., 1984).

During the second phase starting in March 1982, child health interventions were intensified for children under the age of five. The measles vaccine was introduced in half the treatment area and expanded to the other half in November 1985 (Koenig et al., 1990). Additional child health interventions were phased in between 1986 and 1988. In January 1986, DPT (diphtheria, pertussis, and tetanus), polio, and tuberculosis immunizations were added, and, later in that year, vitamin A supplementation. Curative care, such as nutrition rehabilitation for the nutritionally at risk, was introduced in the late 1980s.

During the program roll-out, the comparison area had access to then-standard government health and family planning services. These services included access to modern family planning in the clinics (rather than in the home) but childhood vaccines were not yet available in the comparison area, or most of Bangladesh. Differences between the treatment and comparison areas narrowed after 1988 as lessons of the Matlab success were incorporated into the national plan (Cleland et al., 1994) providing an approximately 10-year experimental period. Starting in 1989, similar vaccines became more readily available nationwide, and the number of government community health workers delivering in-home services increased throughout the country over time. For example, in the comparison area, the worker-

client ratio increased from 1 per 8000 in 1987-1988 to 1 per 5,000 in 1989-1990 in (Cleland et al., 1994), though this ratio was still lower in the treatment area at 1 per 1,300 in 1990.

2.2 MCH-FP Placement

To ensure a rigorous evaluation of the program, treatment and comparison areas were built into the design of the MCH-FP program. Individuals living in the treatment villages were eligible for MCH-FP health and family planning interventions while comparison villages had access to regular government services (Fauveau, 1994). This study area covered about 200,000 people in 149 villages. Villages were not randomly assigned to treatment status; instead, they were allocated to one of four contiguous treatment blocks or to two comparison blocks flanking the treatment area (Figure 1), resulting in a relatively even split into 70 treatment villages and 79 comparison villages. This block design was critical to facilitate access and delivery of program health services and to limit spillovers from vaccination and information on family planning, both which can be substantial (Huber and Khan, 1979). Randomization of villages was not considered practical programatically or for understanding effects from providing vaccinations due to the importance of herd immunity and highly contagious nature of some of the diseases such as measles. The treatment blocks were each organized around a MCH-FP clinic created by the program and staffed by paramedical personal and a male health assistant (Fauveau, 1994). Each comparison area block also had a government-run health clinic. Local female health workers were responsible for conducting in-home visits and were linked to an MCH-FP clinic. The six blocks—four treatment and two comparison—were created specifically for the program and did not correspond to any pre-existing administrative or health service delivery boundaries. The entire study area was fairly homogeneous at the time, but care was taken to assign villages to treatment and comparison blocks in a way that ensured social and economic similarity (Phillips et al., 1982). The area was also generally viewed as geographically insulated from outside influences at the time. Section 5.2 confirms that the placement of villages into treatment and comparison areas indeed resulted in almost all variables being well balanced between the two areas.

2.3 MCH-FP Rollout

The MCH-FP rolled out following the planned timeline detailed in Section 2.1. For example, Figure A1 indicates that the contraceptive prevalence rate (CPR) for married females 15-49 was similarly low prior to the program (<6%) in both areas. There was a large increase in the CPR to 30% in the treatment area during

the first year of the project and it increased steadily, reaching almost 50% by 1988. Due to availability of contraceptives from government services, the CPR did increase in the comparison area over time, though not as quickly, and rates still remained below 20% in 1988.

There was also a rapid rise in the measles vaccination rate to 60% in 1982 after it was introduced in half of the treatment area and in 1985 when it was introduced in the other half (Figure A1). By 1988, coverage rates for children aged 12–23 months living in the treatment area were 93 percent for the BCG vaccine against tuberculosis, 83 percent for all three doses of DPT and polio, 88 percent for measles, and 77 percent for all three major immunizations (Icddr, 2007). In the comparison area, measles vaccination was not regularly provided in government clinics until about 1989, leaving the comparison area a largely unvaccinated population prior to 1989 (Koenig, Fauveau and Wojtyniak, 1991). While no vaccination data exist for the comparison areas during the program rollout, national data confirm that the measles vaccination rate for children under the age of five in Bangladesh was less than 2% in 1986 (Khan and Yoder, 1998). Vaccination rates remained low (below 40%) in the comparison area in 1990 (Fauveau, 1994).

2.4 Effect of the MCH-FP Program on Human Capital in Childhood

Previous literature documents that the MCH-FP program had important effects on the human capital attainment of the first generation in late childhood using survey data from 1996. Children born during the roll-out of the intensive child health intervention (1982-1988) had improved human capital as measured by height (0.22 SD), cognitive functioning (0.39 SD), and years of education (0.17 SD) at ages 8-14 (Barham, 2012). The increase in schooling for these children was concentrated in males (Joshi and Schultz, 2013). There was no improvement in children born before the availability of intensive child health interventions (1977-1981) at ages 15-19 (Barham, 2012). Driessen et al. (2015) demonstrates that improvements in schooling are associated with measles vaccination take-up, consistent with Barham (2012).⁹

3 Conceptual Framework for MCH-FP to Impact Multiple Generations

We specify a conceptual model to illustrate the channels through which the MCH-FP program could have affected the process of human development in adulthood and across generations. We follow Heckman (2007b) and Attanasio, Meghir and Nix (2020) and model the dynamic evolution of human capital, but

9. The program also reduced family size by almost one child (Phillips et al., 1984; Joshi and Schultz, 2013) though the reduction in family size depends on length of exposure of the mother to the program (Barham et al., 2021).

restrict the model to a three-stage framework, encompassing early childhood ($t = 1$), late childhood ($t = 2$), and adulthood ($t = A$). We include a vector of human capital characteristics in the model to capture the multiple dimensions of human capital our analysis considers. The production function for the different dimensions of adult human capital θ_A , can be written as:

$$\theta_A = f(\theta_1, I_1, I_2, \mathbf{Z}, \theta_p) \quad (1)$$

where θ_1 are initial conditions which we simplify to the child's endowment of human capital, I_t represents either public or private investments made during early childhood ($t = 1$) or late childhood ($t = 2$), $\mathbf{Z}$ is a vector of background characteristics (such as gender and religion), and θ_p is a vector of parental human capital. Given conditions outlined in Heckman (2007b), this framework allows for a dynamic human capital production function that embodies qualities of self-reinforcement, dynamic complementarity, and sensitive periods of development.

3.1 Pathways for Impacts on the First Generation

For the first generation, the MCH-FP program represents an increase in public investment, I_1 , in early childhood, as the program provided a package of interventions—vaccinations, family planning, and maternal health—that are known to improve the early childhood health environment. There are five key pathways through which these interventions can enhance adult human capital (θ_A). The first pathway operates through the direct impact of vaccines to reduce both illness and death from infectious diseases. Among these, measles prevention has emerged as particularly crucial due to its long-lasting effects on child health. Recent studies have shown that the measles virus can suppress a child's immune system for up to five years and essentially erases previously developed immunity (Gadroen et al., 2018). This condition, known as “immune amnesia”, severely compromises a child's immune system and has two major consequences: it increases the risk of death from non-measles causes for two years after infection, and it increases the likelihood of illness from non-measles causes for up to five years following infection (Gadroen et al., 2018; Mina et al., 2015).

Second, morbidity from diseases causes general malaise, and apathetic children typically receive less stimulation from adults and less learning opportunities, which may further hinder human capital development (Walker et al., 2007). By improving the disease environment children may benefit from more

stimulation and learning opportunities.

Third, a child's nutritional status is negatively affected by the contagious disease environment, and from secondary complications such as pneumonia and diarrhea (Reddy, 1987). Both non-randomized and randomized studies document that undernutrition—especially before the age of three—impairs the physical growth and cognition of young children. (Grantham-McGregor, Fernald and Sethuraman, 1999*a,b*; Walker et al., 2007). While catch-up growth is possible, it is limited when children experience successive illnesses. Therefore, improving the disease environment can enhance child nutrition, leading to better physical and cognitive development in adulthood.

Fourth, longer birth spacing resulting from the family planning intervention and maternal health inputs may affect the human capital of a child through the improved inutero nutrition (Walker et al., 2007; Almond and Mazumder, 2011).

Lastly, access to modern family planning can lead to a quantity-quality trade-off, with lower-fertility parents investing more in their children both in terms of time spent together and access to resources, such as better nutrition or more schooling which may impact adult human capital measures.

Importantly, it is not a foregone conclusion that an improvement in the early childhood health environment will lead to sustained impacts in adulthood (θ_A). While Barham (2012) shows that the first period investment of MCH-FP was successful at increasing height, cognition and grades attained in the second period, (θ_2), sustained impacts in adult human capital may depend on dynamic complementarity and may require further investment in the second period. The persistence of impacts in adulthood may vary by the type of human capital, and depend on whether period 1 is the only sensitive period for that type of human capital. Investments made in period 1 may also affect adult human capital through epigenetics as the biomedical literature shows one's first period environment can alter gene expression in adulthood.¹⁰

3.2 Pathways for Impacts on the Second Generation

For the second generation, the framework provides several pathways for intergenerational transmission of MCH-FP benefits. First, parental human capital θ_p is directly altered by the MCH-FP program, and this may lead to biological pathways through which higher human capital of the first generation may impact the second generation. The biological pathways are complex and understanding these mechanisms

10. See Gluckman, Hanson and Beedle (2007) and Gluckman, Buklijas and Hanson (2016) for a survey of this literature.

is an active area of research in both human and animal studies. These pathways can be anatomical, physiological, or epigenetic in nature (Drake and Liu, 2010; Fitzgerald, Hor and Drake, 2020; Gluckman, Buklijas and Hanson, 2016). For instance, physical growth of first generation mothers in period 1 could allow them to physically bear larger children, or both parents from the first generation may pass on height and health-determining genes to their children (the second generation) including epigenetic marks resulting from the intervention.

Second, behavioral mechanisms can arise when parental human capital is correlated with the level and productivity of investments made in early and late childhood (I_1 and I_2). Higher human capital parents may be healthier and more able to spend time actively engaging with their children, thus providing a more stimulating environment. Tastes and preferences for health and health-promoting behaviors may also be altered by higher parental human capital. In addition, higher parental human capital could lead to increased income or financial support for investments in children.

Biological and behavioral mechanisms can vary by the sex of the child, making it important to disaggregate effects by sex. There are documented difference by sex in biological mechanisms including nutritional needs, immune response to vaccination, susceptibility to disease, and intergenerational inheritance of traits (Drake and Liu, 2010; Sandovici et al., 2022; Thompson, 2021), with boys more vulnerable to nutritional deficits and infection. On the behavioral side, parental investments may vary by the sex of the child due to persistent environmental factors such as cultural practices, or as a result of program impacts on the first generation outcomes such as their human capital or agency.

This framework also conveys a conceptualization of the interrelation of human capital across three generations. For the first generation, their parents' human capital θ_p , represents their pre-program health endowment, and is a direct input into the their human capital production function. Effects may be heterogeneous depending on the interaction between the parental health endowment and increased public investment (the MCH-FP program) or private investment (I_1). For the second generation, equation 1 predicts that individuals are influenced by their pre-program health endowment, which is the human capital of generation zero (their grandparents). The influence of the pre-program health endowment, however, is mediated through its effect on the human capital of the first generation (their parents), affecting both the health endowment and parental investments of the second generation.

4 Data and Attrition

4.1 Data Sources

The Matlab district has unusually rich panel data owing to it being a demographic surveillance site since 1966 and careful stewardship by icddr,b to ensure use of common identifiers in other datasets. We draw on four main data sources that we link at the individual level using common individual identifiers: the 2012–2015 Matlab Health and Socioeconomic Survey wave 2 (MHSS2), the 1996 Matlab Health and Socioeconomic Survey wave 1 (MHSS1) (Rahman et al., 1999), periodic censuses of the study area conducted by icddr,b in 1974 and 1982 (icddrb, 1974, 1982), and 1974–2014 Matlab demographic surveillance site (DSS) data on vital events (e.g., births, marriages, deaths, in- and out-migrations) collected by icddr,b.¹¹ MHSS1 and MHSS2 are random samples of the study area, while the periodic censuses and DSS data cover the entire study area. The census and DSS data are known for their high quality (collected in the home bi-weekly or monthly) and allow for determination of exact birth dates, treatment status, migration status, and testing of pre-program balance. These data sets are further linked (usually at the village level) to data on potentially confounding programs such as access to microcredit, primary schools, health facilities, flood mitigation, and arsenic exposure. More details on data construction are in [Appendix B](#) and potential confounders in [Appendix C](#).

Outcomes are drawn from MHSS2 which is a large socioeconomic survey designed as a panel follow-up of all individuals in the MHSS1 primary sample and their descendants. The MHSS1 primary sample is representative of the study area's 1996 population, but does not include individuals who migrated between the start of MCH-FP and MHSS1. To address this bias, MHSS2 also includes individuals born to an MHSS1 household member between 1972 and 1989 who had migrated out of Matlab between 1977 and 1996. Finally, MHSS2 sample does not include anyone from households where the whole household and all the descendants migrated between 1977 and 1995, because there was no one left to sample in MHSS1. The annual whole household migration rate from the entire study area over this time period was low, 0.66 percent, and most exiting households were Hindu.

MHSS2 was collected between 2012 and 2015 throughout Bangladesh and included interviews of

11. MHSS2 data linked with the baseline data will be publicly available in the future. It was collected by the authors together with a team of researchers from the University of Colorado Boulder, Brown University, and icddr,b. Specific icddr,b census and DSS data must be requested from the organization. More information can be found at <http://www.icddrb.org/component/content/article/10003-datapolicies/1893-data-policies>

international and difficult-to-track migrants when they returned to the study area to visit family, particularly during Eid celebrations. A phone survey was also administered to international migrants who did not return to Bangladesh during data collection. Phone survey respondents are all men (women rarely migrate internationally) and represent fifteen percent of all men in the sample. In-person survey respondents were interviewed and tested in their homes so the survey does not suffer from attrition that occurs when test measures are collected in institutional settings such as schools. Individual test and health measurements were collected by enumerators who were rigorously trained by psychologists and public health professionals working for icddr.^b While the phone survey is shorter than the main survey, it does include self-reported measures of height and education. However, only one cognitive test was included in the phone survey (digit span forward), so a cognitive index is not available for the phone survey respondents, resulting in smaller sample sizes for cognition for men.

Attrition rates for MHSS2 are less than 10 percent of the target sample which is extremely low even for a survey that is not a 35-year follow-up. Attrition, including death, for the first generation sample is 10 percent for men and 7 percent for women (Appendix Table B1). Second generation sample attrition is less than 7 percent, with attrition from mortality accounting for about 4 percent (Appendix Table B2). Attrition is balanced between the treatment and comparison groups (Appendix Table B3). Appendix B.2 provides more details.

4.2 Outcomes

This paper focuses on four main measures of human capital: height, short stature¹² or stunting, cognition, and grades attained. These outcomes are referred to as stock measures of human capital because they tend to be stable by adulthood. Height and cognition are also used as indicators of early childhood health and nutrition since early childhood is a sensitive period of physical growth and brain development.¹³

To help with interpretation and account for differences in outcomes with age and gender, outcomes are standardized into z-scores by age and sex. We use the WHO international standards for height and stunting of the second generation, otherwise, z-scores are created by internally standardizing by age and sex using the comparison group mean with six-month age bins for those age 6 and younger, and one year age bins for those age 6 and older. Cognition measures are internally standardized into z-scores because

12. Adult short stature is used for the first generation and is defined as height less than 155cm for males or 145cm for females.

13. We do not present effects on metabolic syndrome because its prevalence is still low for this cohort in Bangladesh.

norms for the cognitive tests do not exist for Bangladesh.

MHSS2 includes a battery of cognitive tests covering multiple domains. To reduce concerns of multiple hypothesis testing we create a cognition index by averaging the internally standardized test scores. For the first generation, the cognition index includes the Mini Mental State Exam (MMSE), Digit Span Forwards and Backwards, and Raven's Colored Progressive Matrices (Ravens). Digit Span Forwards and Backwards are tests of working memory which is thought to increase with intelligence; Ravens is a non-verbal and simple measure of general intelligence and perhaps the most common and popular test for people above the age of five (Kaplan and Saccuzzo, 2009); the MMSE was developed as a global assessment of cognitive status and is also a frequently-used, brief cognitive screening test (Ismail, Rajji and Shulman, 2010). Digit Span Forwards is the only cognitive test available in the phone survey so we do not create a cognition index for phone survey respondents, resulting in a lower sample size for men for this outcome. For the second generation, we construct a cognition index using tests of language, fine motor, and gross motor skills which are drawn from the Denver Developmental Screening Test (Frankenburg et al., 1975) for children under age seven. For respondents aged seven and older, the index is comprised of the MMSE, Digit Span Forwards and Backwards, Ravens, a test of memory drawn from a subtest from the Woodcock Johnson III Test of Cognitive Abilities, Memory for Names, and a three minute timed visual matching test to measure perceptual speed. More details on how cognition outcomes are defined are in Appendix B.3.

Finally, we examine effects on grades attained for the first generation, but not the second generation. About half the second generation sample are too young to attend school, and most school age respondents had not yet reached the age of school drop out. Grades attained is taken from individual-level modules in the household survey rather than the testing module where the other outcomes were collected, and the sample size is slightly larger as a result.

4.3 Intent-to-Treat Indicator and Linking to Baseline Variables

MCH-FP eligibility was place-based and dependent on the village of residence. Because a person's residence after the start of the program is potentially endogenous, we use DSS and census data to create an intent-to-treat indicator based on the village of residence of an individual's first household head prior to 1977.¹⁴ Individuals in MHSS1/2 are linked to the DSS and census data by a unique ID available in each

14. The treatment indicator is nearly identical if individuals were linked to 1974 through their fathers and grandfathers. We use the household head to limit missing treatment status and baseline characteristics.

of the datasets using the following sequence of links. First, we link our respondents to the 1974 census through the household head of their first residence in the DSS area. If their household head was not present in the 1974 census, we identify that person's first household head in the DSS area and link that new person to the 1974 census. Finally, remaining unlinked individuals are assigned a treatment status using the location of their household head in the DSS area after the 1974 census, but before the inception of MCH-FP in 1977.¹⁵ The ITT variable, *Treat*, takes a value of one if the 1974 census-linked household head was living in a village in the treatment area in 1974 or migrated into a village in the treatment area between 1974 and 1977, and zero otherwise.

Baseline characteristics from the 1974 census are linked to individuals in the same manner used to construct treatment status. For the few individuals that could not be linked to the 1974 census, missing baseline characteristics are assigned means based on treatment status, sex, and cohort.¹⁶ Finally, the village from the 1974 census link is used to cluster standard errors in our analysis.

5 Empirical Strategy

5.1 Analysis Samples

This paper analyzes the impact of the MCH-FP program on the human capital of two samples—the first and second generation—to determine the longer-term and intergenerational effects, respectively. The first generation sample includes those born during the experimental phase of MCH-FP when both family planning and child health interventions were provided in the treatment area. This cohort, born between 1982–1988 (ages 24–30 in MHSS2), received early childhood investments during a sensitive period of development—corresponding to $t = 1$ in the model. As children older children, they experienced substantial human capital gains from the MCH-FP program, including increases of (Barham, 2012) of 0.22SD in height, 0.17SD in education, and 0.39SD in cognitive functioning, as measured by the Mini Mental State Exam (Barham, 2012).

The second generation sample includes firstborn children of females from the 1982–1988 first-generation cohort.¹⁷ This sample was born between 2002–2015 and are aged 0–14 in MHSS2. We identify

15. Over 96% of sample links to the 1974 census through their first household head and 3 percent link through that person's first household head. The remaining link through their household head's location in the DSS after the 1974 census, but before program inception in October 1977.

16. Only 12 male and 13 female respondents have missing baseline data.

17. We examine effects on children born to fathers in the 1982–1988 cohort because only 34 percent had children. Also, most

children based on pregnancy histories collected in MHSS2 which includes a listing of all live births, still-born children, or lost pregnancies. We follow the children of the females in the first-generation cohort because they are the cohort that experienced human capital gains from the MCH-FP program as children (Barham, 2012), and analyses from this paper shows gains persisted into adulthood. These females are also typically married to older men (mean age 34) who were born prior to the start of intensive child health interventions in 1982 (69% of husbands). The MCH-FP program had limited effects human capital of people born before 1981 on average as teens and young adults (Barham, 2012), so the age difference between spouses helps isolate from which spouse any intergenerational effects on human capital are likely passed.¹⁸ We focus specifically on firstborns because first generation females are early in their fertility: 83 percent of the sample have a firstborn child and 47 percent have a second-born child by the time of the MHSS2 survey. Because mothers with second-born children are more likely to be selected, we exclude these children from the analysis.

While we would ideally examine effects on children born to fathers in the 1982–1988 cohort, only 34 percent of these men had become fathers by the time of data collection. While it is possible that we miss important intergenerational effects by not examining effects on children of first generation fathers, previous research shows intergenerational correlations are stronger between mother and child than father and child (Thompson, 2014; Bhalotra and Rawlings, 2013)

5.2 Pre-program Balance

This paper takes advantage of the treatment and comparison areas built into the design of the MCH-FP program to estimate single-difference ITT effects. We provide five pieces of evidence on pre-program balance to demonstrate the comparison area makes a good counterfactual.

First, since the first generation was born during the program, to provide evidence of pre-program baseline balance in human capital, we examine the human capital of their parents (generation 0) in Table 1 Panel A. This table presents differences in means between the treatment and comparison areas, and normalized differences in means (difference in the means divided by the standard deviation of the mean

married females born after 1988 when interventions were increasingly available in the comparison area, making it difficult to isolate the treatment effects by spouse. Previous research shows intergenerational correlations are stronger between mother and child than father and child (Thompson, 2014; Bhalotra and Rawlings, 2013).

18. MHSS2 did not follow spouses due to cost constraints. However, estimates of program effects on women's spouses education from proxy reports show ITT effects of -0.38 (n=714) which is statistically insignificant.

for the comparison group) that are not influenced by sample size (Imbens and Wooldridge, 2009).¹⁹ Given a lack of pre-program data on human capital for generation-zero parents, we use MHSS1 to examine the balance in human capital for this group for height, cognition as measured by the Mini Mental State Exam, and education. While MHSS1 was collected in 1996 after the program was implemented, the parents of the first-generation sample had mostly completed schooling and growing prior to the start of the program in 1977, so it is unlikely that the program would affect height, education, or cognition. Sample sizes are smaller for the Mini Mental State Exam particularly for fathers because of a data collection error and because it was not collected for many migrations. Table 1 Panel A indicates that generation one's mother's and father's height, years of education, and cognition are balanced between the treatment and comparison area with the differences being small and not statistically different from zero.

Second, to provide further evidence of baseline balance in human capital, we use 1974 pre-program census data on education of the household head and spouse, for baseline households linked to the first generation sample (Section 4.3). There are no other measures of human capital in the census data.²⁰ Table 1 Panel B shows that education was low, but balanced between treatment and comparison areas, in 1974; sixty percent of household heads and 86-87 percent of their spouses had less than 2 years of education in the treatment and comparison group.

Third, to investigate if human capital evolved similarly over time in the treatment and comparison areas, we explore pre-program trends in education. Specifically, we exploit the entire 1974 census (not only households that link to the analysis sample) and split residents born between 1947 and 1968 into 4 or 5 birth year cohorts to examine trends over time in education. We examine cohorts rather than individuals over time because there is only one pre-program data source, the 1974 census. Figure 2 presents the difference in mean education between the treatment and comparison area for each group of birth cohorts. Differences between the treatment and comparison area are close to zero for each of the cohorts indicating there are no pre-program trends over these cohorts. This piece of evidence provides verification that pre-program trends are similar in education.

Fourth, we examine if other baseline individual, household, and household head characteristics, besides education, are balanced in 1974 in Table 1 Panel C. We find pre-program individual and household

19. Normalized differences bigger than 0.25 standard deviations are generally thought to be substantial.

20. We use an indicator variable of if someone has less than 2 years of education to represent amount of education because less than 34 percent of people have more than 2 years of education creating outlier issues. Also, for NGO school we do not know the exact years of schooling attained.

characteristics between the treatment and comparison areas are balanced for the first generation. All normalized differences are small (less than |0.14|) suggesting the differences are not substantial and only 5 out of the 23 baseline characteristics are significantly different below the 5 percent level for the pooled sample.²¹ Characteristics that are significantly different include religion, drinking from a tubewell, latrine ownership, and age of the household head and spouse.

The two largest normalized differences are religion and access to tubewell water. The treatment area had fewer Muslims than the comparison area (84 versus 95 percent of household heads), the remainder being Hindu. All specifications control for if an individual is Muslim and robustness analysis shows that the results remain similar if only Muslims are included in the analysis. In the treatment area, 30 percent of households use tubewells for drinking water compared to only 16 percent of households in the comparison area. The difference in tubewell water access in 1974 is the result of a government program rather than differences in household income, propensity to drill a tubewell, a household's concern about child health, or potentially other unobservables that could be correlated with a person's anthropometrics or cognition. Regardless, the difference in access to tubewell water for drinking is concerning since tubewell water is often thought to be cleaner than other sources of water. However, there is widespread groundwater arsenic contamination in the tubewells in Bangladesh (Chowdhury et al., 2000) which is a serious health concern and has been shown to reduce IQ among school-aged Bangladeshi children (Wasserman et al., 2004). To account for any differences in baseline characteristics we include observables listed in Table 1 Panel B as controls in the regressions, as well as provide robustness analysis to further explore the imbalance between religion and tubewell water.

Lastly, to augment our body of empirical evidence of pre-program balance between the treatment and comparison areas, we lean on prior research that has shown the treatment and comparison areas for the entire Matlab region were balanced prior to the intervention on a number of outcomes related to human capital formation. This includes mortality and fertility (outcomes directly related to the program interventions), but also in migration stocks and flows (Barham, 2012; Barham and Kuhn, 2014; Koenig et al., 1990; Menken and Phillips, 1990).

Taking these five pieces of evidence together reduces concerns that the means of the treatment and comparison areas are similar by chance at baseline, and that unobservables correlated with any of the baseline variables are unbalanced. The evidence strongly suggests that the comparison area is a good

21. Results are similar if examined by sex, though there are only three significant differences.

counterfactual.

5.3 Empirical Specification

We take advantage of the treatment and comparison areas built into the design of the MCH-FP program and estimate single-difference ITT effects of the MCH-FP program on first and second generation human capital. For the first generation, the single-difference model for individual i , from household h , located in village v in 1974, is specified as follows:

$$Y_{ihv} = \beta_0 + \beta_1 \text{Treat}_v + \alpha_t + \mathbf{X}_{ih}\Gamma + \varepsilon_{ihv} \quad (2)$$

where Y_{iv} is the outcome of interest measured in MHSS2, Treat_v is the binary treatment status variable. Birth year fixed effects, α_t , are included to control for differences in the outcome due to year of birth as well as other events that may be correlated with birth year. The vector $\mathbf{X}_{ih}$ controls for individual (sex and religion) and baseline household and household head characteristics in Table 1 panel B. We cluster standard errors at the pre-program village level to account for the placement of villages into treatment and comparison areas and for likely intra-village correlation in the error terms and weight all models to account for attrition from birth to MHSS2 (see Section [Appendix D](#) for details on the weights). Additionally, we report Westfall-Young stepdown adjusted p-values that control for the probability of making any Type I error when testing multiple hypotheses while allowing for dependency among p-values ([Westfall and Young, 1993](#)), and the wild cluster bootstrap at the block level to account for the contiguous nature of villages in the treatment and comparison area and the number of clusters at this level.

In this model, β_1 is the ITT single-difference estimate of the long-term effect of early childhood investments from MCH-FP for those born between 1982–1988 when they are adults aged 24–30 in MHSS2. It represents the combined effect of all program interventions. Identification of causal effects for the single-difference model assumes that the means between the treatment and comparison areas would have been the same in the absence of the program. This is not a testable assumption, but seems reasonable given the evidence presented in Section 5.2 on the pre-program balance in means and pre-program trends. In addition, as robustness we display the p-value from a randomization-based inference test to account for the placement of the treatment area across a contiguous group of villages.²² Furthermore, we discuss

22. With any assignment of village-level treatment status, significant treatment effects could occur simply by chance. Following

and test potential threats to identification in Section 8. For example, we demonstrate there is no evidence of spatially correlated errors, and that other programs (including health and education supply) or changes in the area are driving results. Finally, we present double-difference models using less affected cohorts to control for potential non-time varying differences between treatment and comparison groups and results are similar.

For the second generation, the single-difference model for individual i , born to first-generation mother from village v is specified as follows:

$$Y_{ihv} = \beta_0 + \beta_1 \text{Mother's Treat}_v + \alpha_a + \mathbf{X}_{ih}\Gamma + \varepsilon_{ihv} \quad (3)$$

where Y_{iv} is the outcome of interest, Mother's Treat_v is the binary ITT treatment status variable of the first-generation mother. Six-month age fixed effects, α_a , are included to control for differences in the outcome due to age as well as other events that may be correlated with age. $\mathbf{X}_{ih}$ is the same vector controls of the first-generation mother used in Equation 2. The coefficient β_1 is the single-difference ITT estimator. Similarly to the first generation, we cluster standard errors at the mother's pre-program village level, weight all models to account for attrition from birth to MHSS2 using the mother's weight, and present a similar set of p-values as robustness.

6 First Generation Results

6.1 ITT Effects on Human Capital

We present ITT effects of the MCH-FP program on height measured in cm, an indicator for short stature, a cognition index z-score, and grades attained for the first generation in Table 2 Panel A. The mean of the outcome variable for the comparison group is presented at the bottom of the table. In addition to the standard p-values, we present randomization inference test p-values, Westfall-Young p-values, and the wild cluster bootstrap p-values. Based on the comparison group means in Panel A, 8 percent of men and 11 percent of women are classified as short stature, and average grades attained is low at 7 years for both sexes.

Results pooled by sex demonstrate important ITT effects among the 1982-1988 cohort for height and

Athey and Imbens (2017), we simulate the distribution of treatment effects that would occur from randomly assigning a fixed number of villages to treatment while maintaining the contiguity of treatment and comparison areas.

grades attained, but not for short stature or cognition. This cohort did not experience any impacts as teens and there are still no impacts on this cohort. Specifically, the program led to a 1 cm increase in height (significant at the 1 percent level), and a 0.45 (6 percent) increase in grades attained (significant at the 5 percent level). Effects on short stature are negative but noisy, suggesting there is no distributional impact on height. Effects for cognition are close to zero for the overall index and for each component of the index (Table A1). While growth in height is hard to reverse and educational attainment is necessarily non-decreasing, the positive effects of early childhood investments from MCH-FP on medium-term cognition in 1996 faded out by 2012. The lack of effects in adult cognition could indicate the need for dynamic complementarity through investment in later childhood ($t = 2$), or the difficulty in measuring differences between treatment and comparison areas at an age when cognition is most developed. Earlier program effects may reappear at a later period in the life cycle when cognitive decline commences.

Disaggregating by sex highlights effects are similar between the sexes for height but not for education, and there is no impact on cognition for either sex. For height, males experienced a 0.96 cm increase (significant at the 10 percent level) and females a 1.09 cm increase (significant at the 1 percent level). The standard errors for males may be larger than for females due to measurement error in self-reported height in the phone survey.²³ For education, effects are driven solely by males who experienced 0.97 (14 percent) additional years for grades attained (significant at the 1 percent level).²⁴ The lack of effect on education for females is consistent with the presence of a large nationwide stipend program²⁵ to improve girls' education operating at the time in both the treatment and comparison areas, that is estimated to have increased girls' educational attainment by 1.2 years (Shamsuddin, 2015) (see Appendix C for more details).

6.2 Distributional Effects by the Pre-Program Health Endowment

To understand the effect on the parent-child human capital link, we examine distributional effects by the maternal pre-program human capital endowment (θ_p). As noted in the conceptual framework, the MCH-FP program increased public investment, I_1 , for the first generation. This improvement may impact adult human capital formation differently depending on one's pre-program health endowment. We proxy the pre-program health endowment using generation zero's height, and define a low health endowment

23. Results are similar if the sample is restricted to men not in the phone survey, where height is self-reported.

24. Effects on grades attained are similar to Barham, Kuhn and Turner (2023).

25. This program started in Matlab in 1984 and operated when the first generation was in secondary school.

to be 1 if generation zero is of short stature (less than 145 cm), and zero otherwise. We use height as a proxy because it is unlikely to have been influenced by the MCH-FP program. Adult height serves as a stock measure of health, capturing both genetic factors and conditions experienced during early life. To focus on those with potentially the lowest health endowment, we estimate equation 4 where β_3 compares the treatment effect of individuals with low health endowment (short-stature mothers) to those with a higher health endowment (taller mothers).

$$Y_{ihv} = \beta_0 + \beta_1 \text{Treat}_v \times \text{Low Health Endowment}_i + \beta_2 \text{Treat}_v + \beta_3 \text{Low Health Endowment}_i \quad (4) \\ + \alpha_t + \mathbf{X}_{ih}\Gamma + \varepsilon_{ihv}$$

Results are reported in Table 2 Panel B. Means of the outcome variables for low and higher health endowment individuals are presented at the bottom of the table. They indicate that 26 percent of first generation individuals in the comparison area with a low health endowment are short stature, while only 6 percent of those with a higher health endowment are classified as short-stature. ITT effects for the pooled sample are concentrated in those with the lowest pre-program health endowment. These results demonstrate the program reduced the human capital gap for all outcomes, including stunting and cognition, for which there was no main impact in Panel A. Comparing low to higher health endowment individuals, the ITT effects for the pooled sample show a 1.7 cm increase in height (significant at the 1 percent level), a 15 percentage point (58 percent) reduction in short stature (significant at the 1 percent level), a 0.21 SD increase of in the cognitive function index (significant at the 5 level), and 1 year increase in years education (significant at the 10 percent level). These results suggest that the program generated sizable gains for the initially disadvantaged in the first generation.

Results by sex reveal that program effects are still concentrated among individual with the lowest pre-program health endowment, however, the pattern differs by outcome. For first-generation females, the gap caused by MCH-FP between those with higher versus lower pre-program health endowment are meaningful and statistically significant for height and short stature, with a 2.9 cm increase in height and a 22 percentage point (65 percent) reduction in short stature. In contrast, gaps in cognition and education are larger for males (0.3 SD and 1.2 years) than females (0.08 SD and 0.8 years), though only the gap in cognition for males is statistically significantly (at the 5 percent level), demonstrating that cognition may not have faded since adolescence for everyone.

7 Second Generation Results

7.1 ITT Effects on Human Capital

We examine the intergenerational effects of the program on the human capital of the second generation—firstborn children of first-generation females—who range in age from 0 to 14 in Table 3. Comparison group means for height indicate that, on average, this sample of children exhibits a growth deficit of 1.26 SD, with 26 percent classified as stunted. These growth deficits are relatively similar for males and females, with height deficits of 1.2 and 1.3 standard deviations and stunting rates of 24 and 27 percent, respectively.

Pooled results by sex show no significant intergenerational effects of the program. However, this aggregation masks important heterogeneity by sex. For males, effects are generally small, negative, and statistically insignificant. In contrast, there are sizable and statistically significant effects for firstborn females. Relative to the comparison area, the treatment area shows an intent-to-treat (ITT) effect of a 0.34 standard deviation increase in height-for-age (significant at the 5 percent level), a 14 percentage point—or 50 percent—reduction in stunting (significant at the 1 percent level), and a 0.17 standard deviation improvement in the cognitive index (significant at the 10 percent level). Appendix Table A4 further shows gains across all cognitive domains, except for memory. These findings indicate that the MCH-FP program had meaningful intergenerational impacts on the human capital of second-generation females.

While we would like to mirror the first-generation analysis by examining distributional effects by pre-program health endowment as measured by generation 0's short-stature, the sample sizes are too small. Fewer than 25 children of treatment-group mothers had a short-stature grandmother when disaggregated by sex.

7.2 ITT Effects on Mortality and Fertility

We next examine the effects of the program on first-generation women's fertility and the mortality of their firstborn children, as shown in Table 4. These outcomes are important in their own right, but they also provide insight into whether differential selection into the second generation may be driving the observed intergenerational effects. We estimate single-difference ITT models on all first-generation females in columns 1-6, and a subset of those who have had at least one live birth in columns 7-9. By restricting the sample to mothers, we are conditioning on an endogenous variable, though the vast majority of women

have at least one child (over 83 percent) providing the sample of 677 firstborn children who make up the second generation. We present results for all second-generation children in Panel A and stratified by sex in Panels B and C to investigate whether selection could underlie the differential pattern of effects by sex.

We examine multiple dimensions of fertility and a key mortality measure—the death of the firstborn child. Fertility measures for first-generation women include marital status, age at menarche, whether they had a live birth, number of live births, and whether they experienced a pregnancy loss prior to the birth of their first live child. Intent-to-treat (ITT) effects are close to zero and not statistically significant across all measures for the pooled sample (Panel A), as well as for firstborn males (Panel B) and firstborn females (Panel C). These results suggest that the second-generation findings are not driven by fertility or mortality selection, and that the sex composition of firstborn children is balanced.

7.3 Mechanisms

There are both biological and behavioral pathways along which the MCH-FP program could transmit benefits to the second generation. As outlined in Section 3, from a biological perspective, improvements in parental human capital, θ_p , resulting from the MCH-FP program could potentially influence anatomical, physiological or epigenetic pathways through which MCH-FP impacts the second generation. From a behavioral perspective, parental human capital improvements may affect levels of investments in second generation children (I_1 or I_2), or the productivity of those investments. Identifying which mechanisms are at play is challenging since they may co-vary and their relevance may differ by sex. Furthermore, to rigorously identify the mechanisms, high frequency data and randomization of investments, behaviors, and biological markers is required. However, such data is expensive to collect and experimental variation on mechanisms is rare. While we lack detailed panel data, we are able to examine retrospective information on pertinent early childhood health investments, along with contemporaneous information on key behavioral mechanisms. Below we present analyses of observable mechanisms and provide suggestive evidence that mother's income generation and greater presence in the household may be factors influencing the differing human capital development of second generation girls and boys.

Analysis of mother-level behavior and characteristics in Table 5 reveals that mothers of second-generation daughters in the treatment area are 10 percentage points more likely to report agricultural income and to have savings than those in the comparison area (Table 5, columns 1–4). The increase in agricultural work is from raising animals (ducks and chickens), which takes place within the family compound,

allowing women to care for their children simultaneously. Although the difference in agricultural work between mothers of firstborn females and males is not statistically significant, the magnitude—nearly a 40 percent increase over the male comparison group mean—is notable. The observed increases in agricultural income and savings suggest that mothers of second-generation daughters may have greater monetary and nutritional resources to invest in their children, potentially contributing to the observed sex differences in second-generation outcomes.

We find no further evidence that the MCH-FP affected mother-level variables differently by the sex of their firstborn including household wealth, an index of women's empowerment (see Appendix 4.1 for details on index), or human capital outcomes of first generation mothers.

Figure 3 presents effects on investments in early childhood. For each outcome, the figure displays the point estimate by sex, 95 percent confidence intervals, the comparison group mean (in parentheses), and the p-value testing equality of male and female coefficients. Panel A considers three prenatal and delivery-related investments: (1) whether any pregnancy check-up occurred, (2) whether the birth took place at a skilled facility, and (3) whether delivery was attended by a trained birth attendant. Most health-promoting behaviors show effects near zero and are similar by sex, with the exception that firstborns—regardless of sex—are more likely to be delivered in a skilled facility and attended by a trained assistant. If male and female human capital responds differently to skilled delivery care, with greater gains for females, this could help explain the observed sex-specific pattern of second-generation outcomes.

Figure 3, Panel B, examines three postnatal health investments: the number of vaccinations, the age breastfeeding stopped, and the age the child began eating solid foods. No differential effects are observed, either pooled or by sex. Similarly, there is no impact on preschool attendance, a key measure of early educational investment.

Childcare is typically provided by mothers in this context. Given the high migration rate among first-generation mothers (approximately 30 percent), Figure 3, Panel D, examines whether the mother was the primary caregiver and whether she is currently residing in the household. Mothers of girls are about 10 percent more likely to be currently living household, than boys. While not statistically significant, this suggests greater maternal presence for firstborn girls than boys.

Finally, we examine indicators of a child's positive play both with the parent and on their own. Results are mixed with parents engaging in more positive play with male than female firstborn children, but with firstborn girls engaging in more positive play themselves than boys.

8 Robustness

We examine the robustness of the effects of MCH-FP on the human capital of both generations, including bounding attrition and different weighting schemes, and show the validity of the research design is supported. We present a key subset of robustness tests in Figure 4a and Figure 4b and detailed results for all tests in Appendix Tables A5-A8.

First, given the block nature of the experimental areas, we demonstrate that results are not likely driven by confounding shocks in either the treatment or comparison area. We take advantage of the two distinct geographic areas that form the comparison area—one north and one west (see Figure 1) and repeat the analysis using each comparison block separately in Appendix Tables A5 and A7 Panels B and C. Results remain largely unchanged, though there is some loss of significance due to a smaller sample size. Additionally, we find no evidence of spatially correlated errors across villages (results not shown), nor is there any documented disease outbreak affecting only one area in the decades of demographic surveillance data, even in the early years (Fauveau, 1994). Together, these analyses suggest that neither geographic location nor area-specific shocks are driving the results.

Second, in Appendix Tables A5 and A7 Panel D we include an extended set of controls to account for major changes in Matlab between 1982 and 2012 that could potentially bias the results. Details on available programs and variable construction are provided in Section Appendix C. For the first generation, these controls capture the introduction of a river embankment for irrigation and flood protection in 1987; the expansion of microcredit and other BRAC programs in the 1990s; increased access to primary schools; construction of welfare clinics; availability of modern medical providers and trained midwives; and the discovery of arsenic in some deep tubewells. For the second generation, we control for changes in health and education infrastructure around the time of each child's birth. This includes the Maternal, Neonatal, and Child Health (MNCH) program, launched in 2007, which encouraged facility-based deliveries with skilled obstetric care and was implemented in areas overlapping with the original treatment blocks (Table A7 Panel I). Results remain qualitatively unchanged for both generations, with the exception of the pooled education outcome, where the point estimate is slightly smaller and no longer statistically significant.

Third, we control for a limited set of baseline variables in Panel E that are unbalanced for any sex in either the first or second generation. For both generations, this includes religion, access to tubewell water, and the age and education of the household head and spouse. Again, results are similar.

Fourth, given the imbalance of religion between the treatment and comparison areas prior to MCH-FP, we present the main results for only those that identify as Muslim in Panel F. Again results are similar, though the point estimate on height for females for the second generation is slightly smaller (0.28 SD). Standard errors are also higher due to the smaller sample size.

Fifth, in Panel G we control for any pre-program differences between the treatment and comparison areas using a double-difference model for both generations. For the first generation, we estimate the baseline difference using an older cohort born between 1947-1969. This cohort was born more than eight years prior to the MCH-FP intervention and less likely to have their human capital affected by the program interventions, as shown by the point estimates on the treat variable being close to zero and statistically insignificant. However, they could be selected. Specifically, the migration pattern of this cohort differs by treatment status (Barham and Kuhn, 2014), with migration being lower among the treated than control area. Effects are again similar to the base model, with the exception of there being a larger reduction in short stature for the pooled sample that is marginally significant at the 10 percent level.

For the second generation, identifying a true baseline cohort unaffected by the program is challenging. Instead, we use a less-affected comparison group: children aged 0–14 who are of any birth order but born to women born before 1982—the start of the first-generation cohort. Ideally, we would restrict this group to firstborns to match the treatment group, but the sample size is too small. Double-difference results are qualitatively similar to the single-difference estimates, though the latter appears slightly more conservative based on height outcomes.

Finally, we present results without weights (Panel H of Tables A5 and A7) and adjusting for attrition using Kling-Leibman attrition bounds (Tables A6 and A8).²⁶ Unweighted results are qualitatively and statistically similar, and the bounds show significant results are bounded away from zero and, in most cases, are close to base point estimates. These exercises demonstrate that neither the weighting method nor attrition are driving the results, which is not surprising given the low rates of attrition.

9 Conclusion

We leverage unique data linking three generations and apply a robust causal identification strategy to examine the long-term and intergenerational effects of a widely implemented public health interven-

26. Kling-Leibman bounds assign attritors the mean of the outcome plus or minus 0.10 and 0.25SD for continuous variables, and the worst case scenarios of either all zeros or all ones for binary variables (Kling, Liebman and Katz, 2007).

tion—the MCH-FP program—in Matlab, Bangladesh. Our estimates capture the combined impact of the program’s components, including early child health interventions (e.g., vaccinations), maternal health services, family planning, the service delivery model, and associated externalities. Many of the program interventions continue to form the backbone of preventative public health policy worldwide.

We find important long-term effects of improving the early child health environment on the first generation. This generation experienced increased adult height for both sexes, and grades attained for males of almost a year. However, effects on cognition documented in adolescence (Barham (2012)) faded in early adulthood.

A distinctive contribution of this paper is the examination of how the pre-program health endowment interacted with policy to influence human capital across generations. We find sizable effects for those with the lowest pre-program health endowment across all measures of human capital. While the full sample showed statistically significant effects for only education and height, low-endowment individuals also experienced a 50% reduction in stunting and a 0.2 SD increase in cognition. The finding that the largest gains in adult human capital occurred among individuals with the lowest initial health endowments indicates that, although the MCH-FP program was not specifically targeted at low-endowment individuals, it nonetheless attenuated child–parent human capital correlations and reduced gaps in human capital within the first generation.

Intergenerational effects, following the first born child of mothers of the generation one sample, show effects pass to the next generation. Daughters were taller and 50 percent less likely to be stunted, almost eliminating stunting. Cognitive effects, which faded for the full generation one sample, resurface among the second-generation females. The resurgence of program impacts on cognition in the second generation underscores the importance of considering various dimensions of human capital development. Human capital measures respond differently to one’s environment, and their sensitivity varies across different life stages, making it important to assess effects at different points in the life cycle and across generations.

Mechanisms linking human capital between the generations are not well understood and are difficult and expensive to study. While we are able to rule out that second generation effects are driven simply by mortality or fertility selection resulting from the program and provide suggestive evidence that mother’s employment may play a role, the design of the study did not allow for rigorous identification of behavioral mechanisms. Carefully designed studies and higher-frequency panel data collected during key developmental periods are essential to examine the biological and behavioral mechanisms underlying the

intergenerational transmission of human capital.

The findings of this paper demonstrate that investments in early-life health environments can yield substantial and lasting impacts on human capital. Policymakers who overlook the intergenerational and distributional consequences of such investments risk under-allocating resources for children—thereby missing a critical opportunity to narrow developmental gaps and influence the economic trajectories of future generations.

References

- Almond, Douglas, and Bhashkar Mazumder.** 2011. "Health Capital and the Prenatal Environment: The Effect of Ramadan Observance during Pregnancy." *American Economic Journal: Applied Economics*, 3(4): 56–85.
- Almond, Douglas, Lena Edlund, Hongbin Li, and Junsen Zhang.** 2010. "Long-term effects of early-life development: Evidence from the 1959 to 1961 China famine." In *The economic consequences of demographic change in East Asia*. 321–345. University of Chicago Press.
- Athey, Susan, and Guido W Imbens.** 2017. "The Econometrics of Randomized Experiments." In *Handbook of Economic Field Experiments*. Vol. 1, 73–140. Elsevier.
- Attanasio, Orazio, Costas Meghir, and Emily Nix.** 2020. "Human Capital Development and Parental Investment in India." *The Review of Economic Studies*, 87(6): 2511–2541.
- Attanasio, Orazio P.** 2015. "The Determinants of Human Capital Formation During the Early Years of Life: Theory, Measurement, and Policies." *Journal of the European Economic Association*, 13(6): 949–997.
- Bailey, Martha J.** 2013. "Fifty years of family planning: new evidence on the long-run effects of increasing access to contraception." National Bureau of Economic Research.
- Barham, Tania.** 2012. "Enhancing Cognitive Functioning: Medium-Term Effects of a Health and Family Planning Program in Matlab." *American Economic Journal: Applied Economics*, 4(1): 245–273.
- Barham, Tania, and Randall Kuhn.** 2014. "Staying for Benefits The Effect of a Health and Family Planning Program on Out-Migration Patterns in Bangladesh." *Journal of Human Resources*, 49(4): 982–1013. Publisher: University of Wisconsin Press.
- Barham, Tania, Brachel Champion, Andrew D Foster, Jena D Hamadani, Warren C Jochem, Gisella Kagy, Randall Kuhn, Jane Menken, Abdur Razzaque, Elisabeth Dowling Root, et al.** 2021. "Thirty-five Years Later: Long-term Effects of the Matlab Maternal and Child Health/Family Planning Program on Older Women's Well-being." *Proceedings of the National Academy of Sciences*, 118(28).
- Barham, Tania, Randall Kuhn, and Patrick S Turner.** 2023. "No place like home: Long-run impacts of early child health and family planning on labor and migration outcomes." *Journal of Human Resources*.
- Barr, Andrew, and Chloe Gibbs.** 2017. "Breaking the cycle? Intergenerational effects of an anti-poverty program in early childhood."
- Bartleska, Philipp, Sonja Dobkowitz, Maarit Olkkola, and Michael Rieser.** 2023. "Mass vaccination and educational attainment: Evidence from the 1967–68 Measles Eradication Campaign." *Journal of Health Economics*, 92: 102828.
- Behrman, Jere R, Maria C Calderon, Samuel H Preston, John Hoddinott, Reynaldo Martorell, and Aryeh D Stein.** 2009. "Nutritional Supplementation in Girls Influences the Growth of Their Children: Prospective Study in Guatemala." *The American Journal of Clinical Nutrition*, 90(5): 1372–1379.
- Bhalotra, Sonia, and Samantha Rawlings.** 2013. "Gradients of the Intergenerational Transmission of Health in Developing Countries." *Review of Economics and Statistics*, 95(02): 660–672.
- Bhatia, Shushum, W. H. Mosley, A. S. G. Faruque, and J. Chakraborty.** 1980. "The Matlab Family Planning-Health Services Project." *Studies in Family Planning*, 11(6).
- Black, Sandra, and Paul Devereux.** 2011. "Recent Developments in Intergenerational Mobility." In *Handbook of Labor Economics*. 1487–1541.
- Chowdhury, Uttam K, Bhajan K Biswas, T Roy Chowdhury, Gautam Samanta, Badal K Mandal, Gautam C Basu, Chitta R Chanda, Dilip Lodh, Khitish C Saha, Subhas K Mukherjee, et al.** 2000. "Ground-water Arsenic Contamination in Bangladesh and West Bengal, India." *Environmental Health Perspectives*, 108(5): 393–397.
- Cleland, John, James F Phillips, Sajeda Amin, and Gholam Mostafa Kamal.** 1994. "The determinants of reproductive change in Bangladesh." *Washington, DC: The World Bank*, 72: 1–55.
- Cunha, Flavio, and James Heckman.** 2007. "The Technology of Skill Formation." *American Economic*

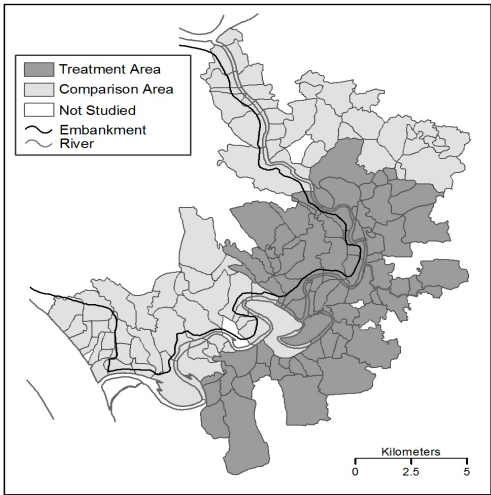
- Review*, 97(2): 31–47.
- Drake, Amanda J, and Lincoln Liu.** 2010. “Intergenerational transmission of programmed effects: public health consequences.” *Trends in Endocrinology & Metabolism*, 21(4): 206–213.
- Driessen, J, A Razzaque, D Walker, and David Canning.** 2015. “The effect of childhood measles vaccination on school enrolment in Matlab, Bangladesh.” *Applied Economics*, 47(55): 6019–6040.
- East, Chloe N, Sarah Miller, Marianne Page, and Laura R Wherry.** 2023. “Multigenerational impacts of childhood access to the safety net: Early life exposure to Medicaid and the next generation’s health.” *American Economic Review*, 113(1): 98–135.
- El Arifeen, Shams, Eva-Charlotte Ekström, Edward A Frongillo, Jena Hamadani, Ashraful I Khan, Ruchira T Naved, Anisur Rahman, Rubhana Raqib, Kathleen M Rasmussen, Katarina Ekholm Selling, et al.** 2018. “Cohort Profile: The Maternal and Infant Nutrition Interventions in Matlab (MINIMat) Cohort in Bangladesh.” *International Journal of Epidemiology*, 47(6): 1737.
- Fauveau, Vincent.** 1994. *Matlab: Women, Children and Health*. Dhaka; icddr,b.
- Fitzgerald, Eamon, Kahyee Hor, and Amanda J Drake.** 2020. “Maternal influences on fetal brain development: The role of nutrition, infection and stress, and the potential for intergenerational consequences.” *Early human development*, 150: 105190.
- Frankenburg, W, J Dodds, AW Fandal, E Kauk, and M Cohrs.** 1975. “The Denver Developmental Screening Test–Revised (DDST).” *Denver, CO: University of Colorado Medical Center*.
- Gadroen, Kartini, Caitlin N Dodd, Gwen MC Masclee, Maria AJ De Ridder, Daniel Weibel, Michael J Mina, Bryan T Grenfell, Miriam CJM Sturkenboom, David AMC Van De Vijver, and Rik L De Swart.** 2018. “Impact and longevity of measles-associated immune suppression: a matched cohort study using data from the THIN general practice database in the UK.” *BMJ Open*, 8(11): e021465.
- Garcia, Jorge Luis, James J. Heckman, and Victor Ronda.** 2021. “The Lasting Effects of early Childhood Education and Promoting the skills and Social Mobility of Disadvantaged African Americans.” *NBER Working Paper No 202183*.
- Gluckman, Peter D, Mark A Hanson, and Alan S Beedle.** 2007. “Early Life Events and Their Consequences for Later Disease: A Life History and Evolutionary Perspective.” *American Journal of Human Biology*, 19(1): 1–19.
- Gluckman, Peter D, Tatjana Buklijas, and Mark A Hanson.** 2016. “The Developmental Origins of Health and Disease (DOHaD) Concept: Past, Present, and Future.” In *The Epigenome and Developmental Origins of Health and Disease*. 1–15. Elsevier.
- Grantham-McGregor, Sally M, Lia C Fernald, and Kavita Sethuraman.** 1999a. “Effects of Health and Nutrition on Cognitive and Behavioural Development in Children in the First Three Years of Life: Part 1: Low Birthweight, Breastfeeding, and Protein-Energy Malnutrition.” *Food and Nutrition Bulletin*, 20(1): 53–75.
- Grantham-McGregor, Sally M, Lia C Fernald, and Kavita Sethuraman.** 1999b. “Effects of Health and Nutrition on Cognitive and Behavioural Development in Children in the First Three Years of Life: Part 2: Infections and Micronutrient Deficiencies: Iodine, Iron, and Zinc.” *Food and Nutrition Bulletin*, 20(1): 76–99.
- Grönqvist, Erik, Björn Öckert, and Jonas Vlachos.** 2017. “The Intergenerational Transmission of Cognitive and Noncognitive Abilities.” *Journal of Human Resources*, 52(4): 887–918.
- Heckman, James J.** 2007a. “The economics, technology, and neuroscience of human capability formation.” *Proceedings of the National Academy of Sciences*, 104(33): 13250.
- Heckman, James J.** 2007b. “The Economics, Technology, and Neuroscience of Human Capability Formation.” *Proceedings of the National Academy of Sciences*, 104(33): 13250–13255.
- Hertz, Tom, Tamara Jayasundera, Patrizio Piraino, Sibel Selcuk, Nicole Smith, and Alina Verashchagina.** 2008. “The Inheritance of Educational Inequality: International Comparisons and Fifty-Year Trends.” *The B.E. Journal of Economic Analysis & Policy*, 7(2).

- Hoynes, Hilary, Diane Whitmore Schanzenbach, and Douglas Almond.** 2016. "Long-Run Impacts of Childhood Access to the Safety Net." *American Economic Review*, 106(4): 903–34.
- Huber, Douglas H, and Atiqur Rahman Khan.** 1979. "Contraceptive Distribution in Bangladesh Villages: the Initial Impact." *Studies in Family Planning*, 10(8/9): 246–253.
- icddr.** 1974. "Matlab 1974 Census."
- icddr.** 1982. "Matlab 1982 Census."
- Icddr, B.** 2007. "Health and demographic surveillance system-Matlab." *2005 Socioeconomic Census*.
- Imbens, Guido W, and Jeffrey M Wooldridge.** 2009. "Recent Developments in the Econometrics of Program Evaluation." *Journal of Economic Literature*, 47(1): 5–86.
- Ismail, Zahinoor, Tarek K Rajji, and Kenneth I Shulman.** 2010. "Brief Cognitive Screening Instruments: An Update." *International Journal of Geriatric Psychiatry*, 25(2): 111–120.
- Joshi, Shareen, and T Paul Schultz.** 2013. "Family Planning and Women's and Children's Health: Long-term Consequences of an Outreach Program in Matlab, Bangladesh." *Demography*, 50(1): 149–180.
- Kaplan, RM, and DP Saccuzzo.** 2009. "Standardized Tests in Education, Civil Service, and the Military." *Psychological testing: Principles, Applications, and Issues*, 7: 325–327.
- Khan, M Mahmud, and Richard Yoder.** 1998. *Expanded Program on Immunization in Bangladesh: Cost, Cost-effectiveness, and Financing Estimates*. Bethesda, MD: Partnerships for Health Reform Project, Abt Associates Inc., 1998.
- Kling, Jeffrey R, Jeffrey B Liebman, and Lawrence F Katz.** 2007. "Experimental Analysis of Neighborhood Effects." *Econometrica*, 75(1): 83–119.
- Koenig, M. A., M. A. Khan, B. Wojtyniak, J. D. Clemens, J. Chakraborty, V. Fauveau, J. F. Phillips, J. Akbar, and U. S. Barua.** 1990. "Impact of Measles Vaccination on Childhood Mortality in Rural Bangladesh." *Bulletin of the World Health Organization*, 68(4): 441–447.
- Koenig, Michael A., Vincent Fauveau, and Bogdan Wojtyniak.** 1991. "Mortality Reductions from Health Interventions: The Case of Immunization in Bangladesh." *Population and Development Review*, 17(1).
- Maluccio, John A, John Hoddinott, Jere R Behrman, Reynaldo Martorell, Agnes R Quisumbing, and Aryeh D Stein.** 2009. "The Impact of Improving Nutrition During Early Childhood on Education Among Guatemalan Adults." *The Economic Journal*, 119(537): 734–763.
- Menken, Jane, and James F Phillips.** 1990. "Population Change in a Rural Area of Bangladesh, 1967–87." *The Annals of the American Academy of Political and Social Science*, 510(1): 87–101.
- Miller, Sarah, and Laura R Wherry.** 2019. "The Long-Term Effects of Early Life Medicaid Coverage." *Journal of Human Resources*, 54(3): 785–824.
- Mina, Michael J, C Jessica E Metcalf, Rik L De Swart, ADME Osterhaus, and Bryan T Grenfell.** 2015. "Long-Term Measles-Induced Immunomodulation Increases Overall Childhood Infectious Disease Mortality." *Science*, 348(6235): 694–699.
- Noghanibehambari, Hamid.** 2024. "Intergenerational Benefits of Childhood Health Intervention." *American Journal of Health Economics*, 10(4): 670–698.
- Painter, R. C., C. Osmond, P. Gluckman, M. Hanson, D. I. W. Phillips, and T. J. Roseboom.** 2008. "Trans-generational effects of prenatal exposure to the Dutch famine on neonatal adiposity and health in later life." *BJOG: An International Journal of Obstetrics & Gynaecology*, 115(10).
- Phillips, James F, Ruth Simmons, J. Chakraborty, and A. I. Chowdhury.** 1984. "Integrating Health Services into an MCH-FP Program: Lessons from Matlab, Bangladesh." *Studies in Family Planning*, 15(4).
- Phillips, James F, Wayne S Stinson, Shushum Bhatia, Makhlisur Rahman, and Jyotsnamoy Chakraborty.** 1982. "The Demographic Impact of the Family Planning–Health Services Project in Matlab, Bangladesh." *Studies in Family Planning*, 131–140.
- Rahman, Anisur, Allisyn Moran, Jesmin Pervin, Aminur Rahman, Monjur Rahman, Sharifa Yeasmin, Hosnara Begum, Harunor Rashid, Mohammad Yunus, Daniel Hruschka, et al.** 2011. "Effectiveness of an Integrated Approach To Reduce Perinatal Mortality: Recent Experiences From Matlab, Bangladesh."

- BMC Public Health*, 11(1): 1–14.
- Rahman, O, J Menken, A Foster, C Peterson, MN Khan, R Kuhn, and P Gertler.** 1999. “The Matlab Health and Socio-economic Survey: Overview and User’s Guide, 1999.”
- Reddy, Vinodini.** 1987. “Interaction Between Nutrition and Measles.” *The Indian Journal of Pediatrics*, 54(1): 53–57.
- Richter, André, and Per Olof Robling.** 2013. “Multigenerational Effects of the 1918-19 Influenza Pandemic in Sweden.” *Working paper / Swedish Institute for Social Research (SOFI)*.
- Sandovici, Ionel, Denise S Fernandez-Twinn, Antonia Hufnagel, Miguel Constância, and Susan E Ozanne.** 2022. “Sex differences in the intergenerational inheritance of metabolic traits.” *Nature Metabolism*, 4(5): 507–523.
- Shamsuddin, Mrittika.** 2015. “Labour Market Effects of a Female Stipend Programme in Bangladesh.” *Oxford Development Studies*, 43(4): 425–447.
- Thompson, Amanda L.** 2021. “Greater male vulnerability to stunting? Evaluating sex differences in growth, pathways and biocultural mechanisms.” *Annals of human biology*, 48(6): 466–473.
- Thompson, Owen.** 2014. “Genetic Mechanisms in the Intergenerational Transmission of Health.” *Journal of Health Economics*, 35: 132–146.
- van Ginneken, Jeroen, Radheshyam Bairagi, Andres de Francisco, Asdul Mazid Sardar, and Patrick Vaughan.** 1998. *Health and demographic surveillance in Matlab: past, present and future*. International Centre for Diarrhoeal Disease Research, Bangladesh.
- Venkataramani, Atheendar S.** 2011. “The Intergenerational Transmission of Height: Evidence From Rural Vietnam.” *Health Economics*, 20(12): 1448–1467.
- Walker, Michael W, Alice H Huang, Suleiman Asman, Sarah J Baird, Lia Fernald, Joan Hamory Hicks, Fernando Hoces de la Guardia, Satoshi Koiso, Michael Kremer, Matthew N Krupoff, et al.** 2023. “Intergenerational child mortality impacts of deworming: Experimental evidence from two decades of the Kenya Life Panel Survey.” National Bureau of Economic Research.
- Walker, Susan P, Susan M Chang, Amika S Wright, Rodrigo Pinto, James J Heckman, and Sally M Grantham-McGregor.** 2021. “Cognitive, Psychosocial, and Behaviour Gains at Age 31 Years From the Jamaica Early Childhood Stimulation Trial.” *Journal of Child Psychology and Psychiatry*.
- Walker, Susan P, Theodore D Wachs, Julie Meeks Gardner, Betsy Lozoff, Gail A Wasserman, Ernesto Pol-litt, Julie A Carter, International Child Development Steering Group, et al.** 2007. “Child Development: Risk Factors for Adverse Outcomes in Developing Countries.” *The Lancet*, 369(9556): 145–157.
- Wasserman, Gail A, Xinhua Liu, Faruque Parvez, Habibul Ahsan, Pam Factor-Litvak, Alexander van Geen, Vesna Slavkovich, Nancy J Lolacono, Zhongqi Cheng, Iftikhar Hussain, et al.** 2004. “Water Arsenic Exposure and children’s Intellectual Function in Araihaazar, Bangladesh.” *Environmental Health Perspectives*, 112(13): 1329–1333.
- Westfall, Peter H, and S Stanley Young.** 1993. *Resampling-based multiple testing: Examples and methods for p-value adjustment*. Vol. 279, John Wiley & Sons.

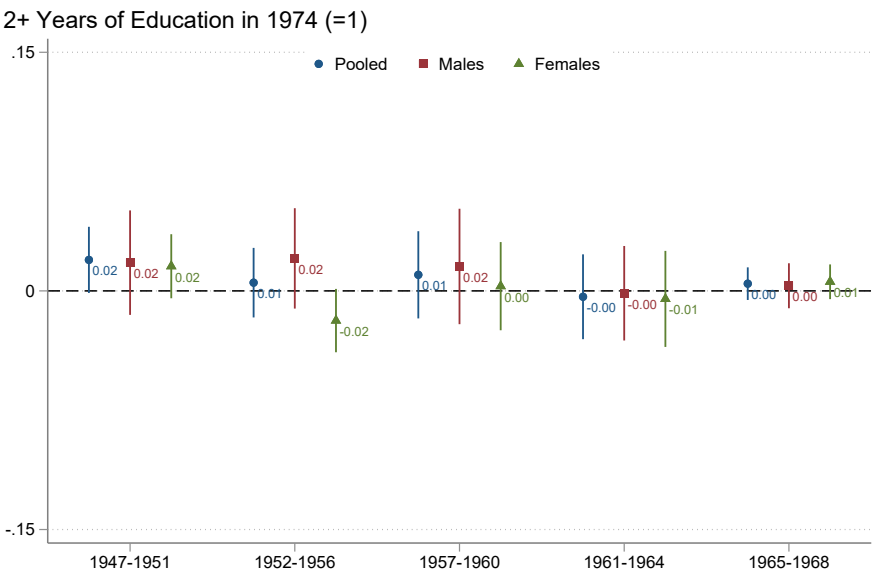
Figures and Tables

FIGURE 1 — MAP OF MATLAB STUDY AREA



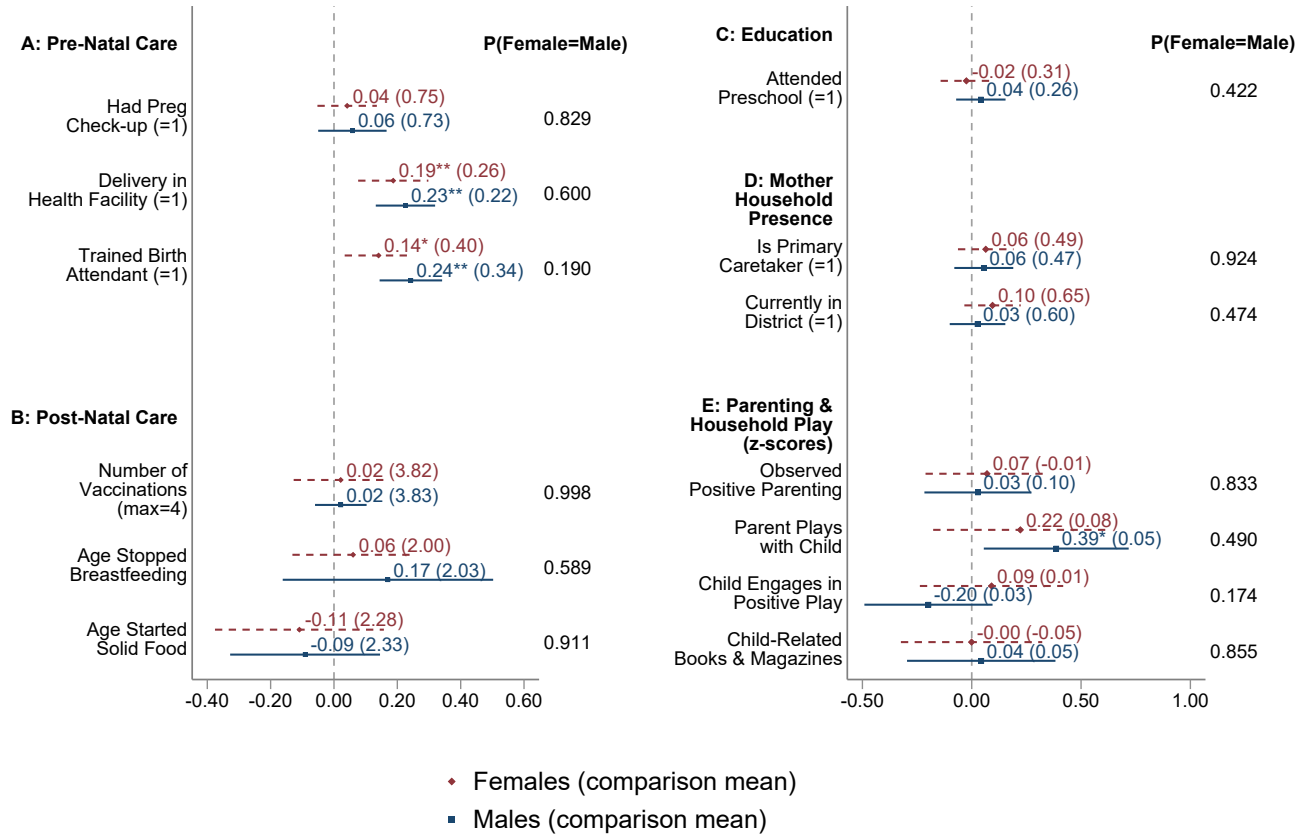
Notes: This figure shows the Matlab study site in Bangladesh where the MCH-FP program was implemented. Subdivisions indicate distinct villages where an individual's household head lived in 1974 used to create the intent-to-treat measure.

FIGURE 2 — TRENDS IN PRE-PROGRAM EDUCATION



Notes: This figure shows trends in pre-program education differences between treatment and comparison areas with 95% confidence intervals. Education data is taken from the 1974 census conducted prior to MCH-FP start in 1977 and dichotomized into less than two or two or more years of education completed. All treatment-comparison area difference estimates include birth year fixed effects, individual and pre-program household characteristics listed in Table 1 Panel B.

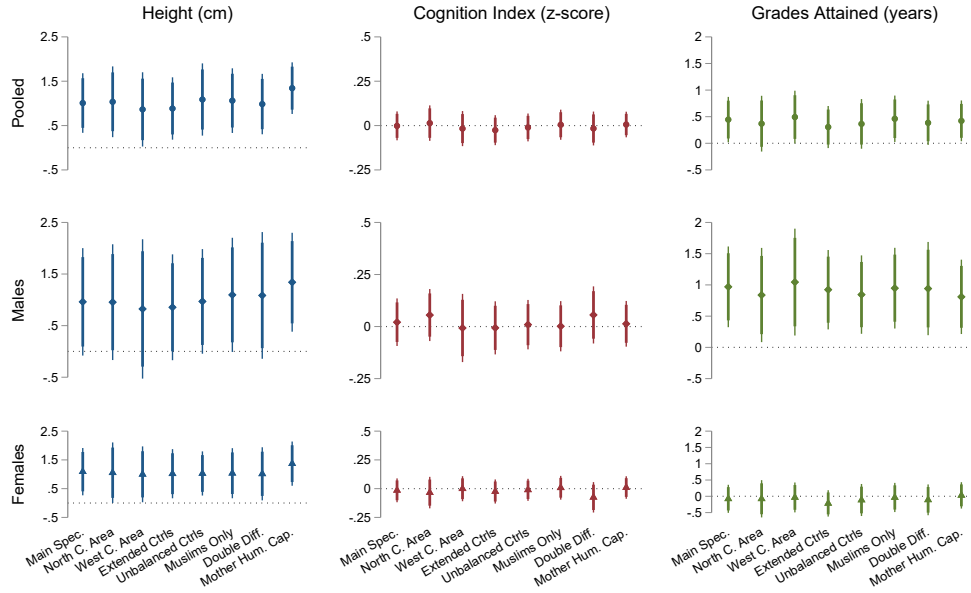
FIGURE 3 — SECOND GENERATION: POTENTIAL MECHANISMS



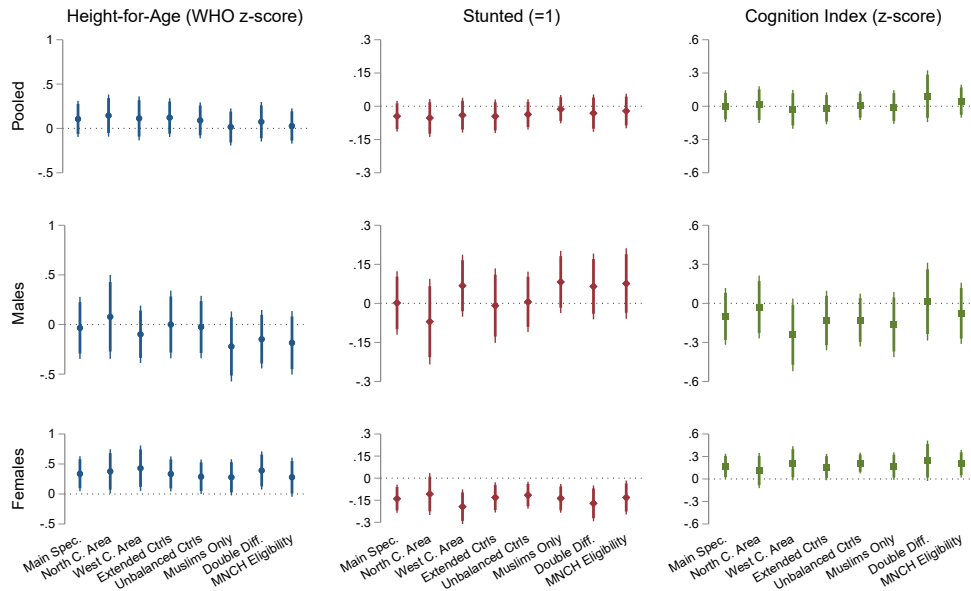
Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This figure shows the ITT effects for firstborn children born to 1982-88 mothers from Equation 3, separately by males and females. The mean of the comparison group is displayed in parentheses and the bars represent 95% confidence intervals. *Delivery in Health Facility* equals one if the pregnancy was delivered in a certified medical center. *Trained Birth Attendant* equals one if the delivery was performed by a physician, nurse or other healthcare worker. *Number of Vaccinations* is the total types of vaccinations received out of four (BCG, Polio, DPT and MMR). *Parenting & Household Play z-scores* are internally-standardized indexes of the interviewer's observation of various parent-child interactions. See Appendix B for a fuller description of each potential mechanism.

FIGURE 4 — SELECTED ROBUSTNESS CHECKS

A: First Generation



B: Second Generation



Notes: This figure shows ITT effects on human capital for both generations are robust to alternate specifications. Thinner bars represent the 95% confidence interval while thicker bars represent the 90% confidence interval. *Main Spec* shows our preferred model for context. *North C. Area* and *West C. Area* restrict the sample of comparison units to those from the northern and western comparison areas, respectively (see Figure 1). *Extended Ctrl.* includes additional controls to account for changes over time in the supply of health and education. *Unbalanced Ctrl.* limits the pre-program baseline control set to only those that are unbalanced. *Muslims Only* limits the sample to Muslims which are over represented in the comparison area. *Double Diff.* estimates the ITT effect of MCH-FP with a double difference model using a less-affected cohort. *Mother Hum. Cap.* includes mother's 1996 height and years of education. *MNCH Eligibility* adds a control for the 2007 Maternal, Neonatal and Child Health Project (MNCH). See Section 8 and Appendix C for a full description of each model and Appendix A for additional robustness checks.

TABLE 1 — FIRST GENERATION: TREATMENT AND COMPARISON AREA BALANCE

	Treatment Area			Comparison Area			Difference		
	Mean	SD	Obs	Mean	SD	Obs	Mean	T-stat	Mean/SD
<i>Panel A: Gen 1 Parents' (Gen 0) Human Capital in 1996 MHSS1</i>									
Mother's height (cm)	149.49	9.13	702	150.11	7.81	847	-0.62	-1.41	-0.05
Father's height (cm)	160.96	7.93	616	160.95	8.89	767	0.01	0.02	0.00
Mother's MMSE (z-score)	-0.40	1.54	650	-0.44	1.43	785	0.04	0.50	0.02
Father's MMSE (z-score)	-0.44	1.14	508	-0.55	1.62	605	0.11	1.34	0.06
Mother's education (years)	1.71	3.41	705	1.58	3.64	851	0.13	0.73	0.03
Father's education (years)	3.50	3.90	691	3.20	5.00	837	0.29	1.29	0.05
<i>Panel B: Household Head (HH) and Spouse Education in 1974 Census</i>									
HH <2 years of education (=1)	0.60	0.56	742	0.61	0.78	896	-0.01	-0.34	-0.01
HH spouse <2 years of education (=1)	0.86	0.43	742	0.87	0.48	896	-0.01	-0.59	-0.02
<i>Panel C: Individual and Household Characteristics in 1974 Census</i>									
Male (=1)	0.51	0.46	742	0.50	0.46	896	0.01	0.61	0.02
Birth year	1984.97	1.76	742	1984.93	1.89	896	0.03	0.38	0.01
Muslim (=1)	0.84	0.81	742	0.95	0.31	896	-0.12	-3.81	-0.14
Bari size	8.64	10.26	742	8.03	10.40	896	0.61	1.19	0.04
Family size	6.79	3.51	742	6.54	3.08	896	0.25	1.53	0.05
Latrine (=1)	0.81	0.67	742	0.88	0.73	896	-0.07	-2.03	-0.07
Owens a lamp (=1)	0.62	0.65	742	0.59	0.74	896	0.03	0.91	0.03
Owens a watch (=1)	0.14	0.42	742	0.15	0.45	896	-0.01	-0.41	-0.01
Owens a radio (=1)	0.08	0.34	742	0.07	0.28	896	0.01	0.34	0.01
Number of cows	1.41	2.25	742	1.33	1.98	896	0.08	0.71	0.03
Number of boats	0.65	0.95	742	0.65	0.99	896	-0.01	-0.15	-0.01
Wall tin or tinmix (=1)	0.29	0.56	742	0.29	0.60	896	-0.01	-0.22	-0.01
Tin roof (=1)	0.82	0.50	742	0.83	0.54	896	-0.01	-0.41	-0.01
Number of rooms per capita	0.22	0.11	742	0.22	0.12	896	0.00	0.79	0.03
Drinking water, tubewell (=1)	0.30	0.88	742	0.16	0.74	896	0.15	3.59	0.13
Drinking water, tank (=1)	0.40	1.07	742	0.33	1.31	896	0.07	1.12	0.04
HH age	48.21	16.83	742	45.94	16.36	896	2.27	2.75	0.10
HH works in agriculture (=1)	0.61	0.67	742	0.56	0.73	896	0.04	1.22	0.04
HH works in fishing (=1)	0.07	0.47	742	0.07	0.35	896	0.00	-0.08	0.00
HH works in business (=1)	0.10	0.32	742	0.10	0.41	896	0.00	0.07	0.00
HH spouse's age	37.62	15.65	742	35.69	15.05	896	1.93	2.53	0.09
1982 Land size	9.99	19.29	742	10.96	19.79	896	-0.97	-1.00	-0.04

Notes: This table shows balance of pre-program individual, parental and household characteristics between treatment and comparison areas. Panel A shows the human capital of mothers and fathers in 1996 using MHSS1, filling in missing values of height and cognition with 2012 values (MHSS2) and of education with their spouse's report in 1996 (MHSS1). Panel B shows educational attainment using the 1974 census of household heads of respondents who have education data in MHSS2. Panel C displays respondents' individual characteristics and characteristics of their pre-program household or household head (HH). Household characteristics are drawn from the 1974 census with the exception of land size which was only available in the 1982 census. Standard deviations (SD) are clustered at the pre-program village level. Household head and spouse age are reported, but these variables are likely affected by the family planning program which increased birth intervals and decreased family sizes in the treatment area (Barham et al., 2021). Observations are weighted to correct for attrition between birth and the 2012 MHSS2 survey.

TABLE 2 — FIRST GENERATION: ITT EFFECTS ON HUMAN CAPITAL

	Pooled				Males				Females			
	Height (cm)	Short Stature (=1)	Cognition Index (z-score)	Grades Attained (years)	Height (cm)	Short Stature (=1)	Cognition Index (z-score)	Grades Attained (years)	Height (cm)	Short Stature (=1)	Cognition Index (z-score)	Grades Attained (years)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
<i>Panel A: Average Effects</i>												
Treat	1.008 (0.340)**	-0.018 (0.017)	-0.002 (0.041)	0.446 (0.216)*	0.961 (0.527)+	-0.008 (0.021)	0.021 (0.058)	0.969 (0.327)**	1.090 (0.416)**	-0.025 (0.026)	-0.015 (0.052)	-0.077 (0.218)
P-value Adjustments												
Rand. Inf.	0.00	0.21	0.99	0.07	0.01	0.12	0.13	0.00	0.00	0.05	0.14	0.14
Westfall-Young	0.00	0.27	0.79	0.01	0.04	0.86	0.99	0.00	0.00	0.49	0.88	0.88
Wild Cluster Bootstrap	0.03	0.50	0.94	0.03	0.09	0.75	0.50	0.00	0.03	0.47	0.53	0.69
Mean of Comparison	157.32	0.09		7.11	163.40	0.08		7.00	151.31	0.11		7.23
Obs	1633	1633	1496	1638	828	828	692	831	805	805	804	807
<i>Panel B: Heterogeneity by Health Endowment</i>												
Treat × Low Health Endowment	1.706 (0.629)**	-0.152 (0.049)**	0.206 (0.097)*	1.023 (0.515)*	0.846 (1.009)	-0.078 (0.062)	0.309 (0.141)*	1.158 (0.719)	2.891 (0.837)**	-0.224 (0.067)**	0.081 (0.143)	0.800 (0.651)
Treat	0.891 (0.343)*	0.008 (0.017)	-0.021 (0.047)	0.315 (0.235)	1.003 (0.517)+	0.009 (0.020)	-0.006 (0.067)	0.860 (0.362)*	0.692 (0.422)	0.013 (0.026)	-0.028 (0.056)	-0.227 (0.220)
Low Health Endowment	-4.780 (0.417)**	0.198 (0.034)**	-0.184 (0.069)**	-0.650 (0.311)*	-4.377 (0.701)**	0.138 (0.044)**	-0.165 (0.100)+	-0.582 (0.507)	-5.433 (0.624)**	0.253 (0.050)**	-0.202 (0.101)*	-0.764 (0.335)*
Means of Comparison												
Low Health Endowment	152.99	0.26	0.26	6.41	160.02	0.18	0.24	6.64	146.64	0.34	0.27	6.20
High Health Endowment	158.30	0.06	0.47	7.29	164.15	0.05	0.37	7.05	152.34	0.06	0.57	7.54
Obs	1544	1544	1412	1549	783	783	652	786	761	761	760	763

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects on the stock of human capital of the first generation. Standard errors are clustered at the pre-program village level and reported in parentheses. Random inference p-values are calculated from a distribution of test statistics constructed by randomly reassigning treatment status to villages over 10,000 permutations while maintaining geographic contiguity. Westfall-Young p-values are adjusted for the probability of any Type I error when testing multiple hypotheses while allowing for dependency among p-values. Wild cluster bootstrap p-values are clustered at the six-block level. All regressions include individual and pre-intervention characteristics interacted with birth cohort and are weighted to correct for attrition between birth and the 2012 MHSS2 survey. Individual characteristics include year of birth fixed effects, age cohort fixed effects, and religion. Pre-intervention characteristics include all characteristics in Table 1. Short stature is defined as height less than 155cm for males and 145cm for females. The Cognition Index includes MMSE, Digit Spans Forward and Backward, and Ravens Progressive Matrix scores. Male Cognition Index sample sizes are smaller than female because tests were not administered in the phone survey.

TABLE 3 — SECOND GENERATION: ITT EFFECTS ON HUMAN CAPITAL

	Pooled			Males			Females		
	Height- for-Age (WHO z-score) (1)	Stunted (=1) (2)	Cognition Index (z-score) (3)	Height- for-Age (WHO z-score) (4)	Stunted (=1) (5)	Cognition Index (z-score) (6)	Height- for-Age (WHO z-score) (7)	Stunted (=1) (8)	Cognition Index (z-score) (9)
Mother's Treat	0.106 (0.103)	-0.045 (0.035)	0.001 (0.073)	-0.034 (0.158)	0.002 (0.061)	-0.100 (0.110)	0.337 (0.147)*	-0.140 (0.048)**	0.166 (0.086)+
P-value Adjustments									
Rand. Inf. P-value	0.243	0.192	0.975	0.867	0.974	0.481	0.008	0.072	0.068
Westfall-Young	0.317	0.252	0.984	0.970	0.970	0.525	0.017	0.004	0.028
Wild Cluster Bootstrap	0.062	0.156	0.938	0.875	0.969	0.344	0.000	0.000	0.094
Mean	-1.262	0.258		-1.221	0.242		-1.302	0.273	
Obs	629	629	596	311	311	294	318	318	302

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects on the stock of human capital of second generation children of females born 1982-88 using their mother's treatment status. Standard errors in parentheses are clustered at the mother's pre-program village level. Random inference p-values are calculated from a distribution of test statistics constructed by randomly reassigning mother's treatment status to villages over 10,000 permutations while maintaining geographic contiguity. Westfall-Young p-values are adjusted for the probability of any Type I error when testing multiple hypotheses while allowing for dependency among p-values. Wild cluster bootstrap p-values are clustered at mother's six block level. Means are for children of comparison group mothers. All regressions include child's six-month age fixed effects, mother's year of birth fixed effects, mother's individual and pre-intervention characteristics listed in Table A3 and are weighted to correct for attrition between birth and the 2012 MHSS2 survey. Height-for-age is calculated using WHO standards. Stunted equals one if the child's height-for-age is less than 2 SD below the mean. Cognition index is internally standardized by sex and age using the comparison group's mean and standard deviation. For ages 0-6, the cognition index is the average of the standardized language, fine motor, and gross motor Denver scores. For ages 7-14, it is the average of standardized MMSE, memory, digit spans, Ravens, and matching test scores.

TABLE 4 — FIRST GENERATION FEMALES: ITT EFFECTS ON FERTILITY AND CHILD MORTALITY BY SEX OF THE FIRSTBORN CHILD

	All Females					Females with ≥ 1 Live Birth			
	Fertility					Fertility			
	Married (=1)	Age at Menarche (years)	Any Live Birth (=1)	Number of Live Births	Lost Preg- nancy Before Firstborn (=1)	Child Mortality Firstborn Later Died (=1)	Male Firstborn (=1)	Age at First Birth (years)	Has Secondborn (=1)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Panel A: All First Generation Females</i>									
Treat	0.002 (0.017)	-0.056 (0.072)	0.015 (0.025)	-0.039 (0.065)	-0.019 (0.021)	-0.014 (0.012)	-0.006 (0.043)	-0.068 (0.246)	-0.018 (0.038)
Mean	0.933	13.462	0.838	1.516	0.083	0.045	0.505	21.535	0.559
Obs	817	805	811	811	811	811	677	677	677
<i>Panel B: First Generation Mothers With Male Firstborn</i>									
Treat		-0.081 (0.101)		-0.008 (0.078)		-0.012 (0.026)		-0.174 (0.343)	0.043 (0.056)
Mean		13.428		1.766		0.080		21.612	0.548
Obs		340		341		341		341	341
<i>Panel C: First Generation Mothers With Female Firstborn</i>									
Treat		-0.011 (0.127)		-0.111 (0.109)		-0.008 (0.017)		-0.040 (0.392)	-0.058 (0.064)
Mean		13.429		1.853		0.027		21.457	0.571
Obs		334		336		336		336	336
P(Male=Female)		0.658		0.458		0.887		0.790	0.249

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects on marriage, fertility and child mortality outcomes of first generation females. Standard errors are clustered at the pre-program village level. All regressions include individual characteristics and pre-intervention characteristics interacted with birth cohort and are weighted to correct for attrition between birth and the 2012 MHSS2 survey. Individual characteristics include year of birth fixed effects and religion. Pre-intervention characteristics include all individual and household characteristics listed in Table 1.

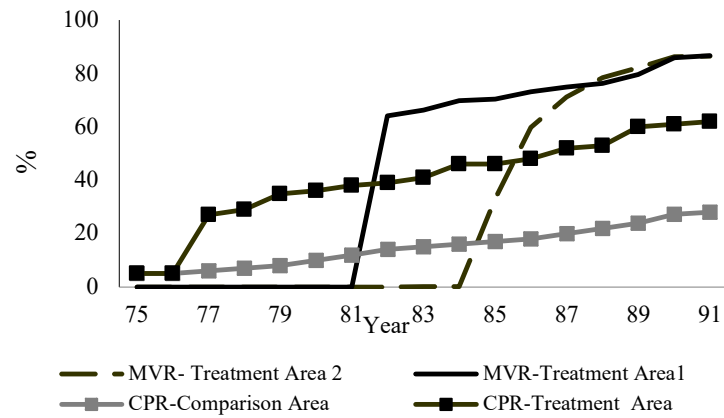
TABLE 5 — FIRST GENERATION FEMALES: ITT EFFECTS ON MOTHER’S BEHAVIORAL OUTCOMES BY SEX OF THE FIRSTBORN CHILD

	Individual Income, Assets, & Savings					Household Wealth (2012 USD)		Empowerment	Human Capital		
	Has Labor	Has Agricultural	Raises	Has	Has Micro-	Assets	Consumption	(0-1)	Height	Cognition	Grades
	Income	Income	Animals	Savings	credit Loan				(cm)	Index	Attained
	(=1)	(=1)	(=1)	(=1)	(=1)				(z-score)	(years)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
<i>Panel A: All First Generation Females</i>											
Treat	0.065	0.063	0.063	0.086	0.057	-177.046	19.556	0.005	1.040	0.016	0.014
	(0.032)*	(0.024)*	(0.024)*	(0.033)*	(0.036)	(221.380)	(35.606)	(0.012)	(0.443)*	(0.054)	(0.218)
Mean	0.255	0.153	0.153	0.201	0.237	1880.524	666.252	0.364	151.398	0.462	6.957
Obs	677	677	677	673	676	677	677	676	673	673	675
<i>Panel B: First Generation Mothers With Male Firstborn</i>											
Treat	0.067	0.045	0.045	0.073	0.036	-264.100	-18.010	0.011	1.205	-0.037	-0.233
	(0.045)	(0.040)	(0.040)	(0.045)	(0.045)	(416.685)	(51.529)	(0.017)	(0.601)*	(0.092)	(0.307)
Mean	0.266	0.165	0.165	0.193	0.251	1793.964	660.400	0.358	151.079	0.432	6.941
Obs	341	341	341	339	340	341	341	340	340	340	340
<i>Panel C: First Generation Mothers With Female Firstborn</i>											
Treat	0.057	0.102	0.102	0.098	0.089	-251.852	25.133	-0.001	1.193	0.047	0.230
	(0.045)	(0.040)*	(0.040)*	(0.047)*	(0.054)	(220.396)	(49.926)	(0.016)	(0.598)*	(0.079)	(0.381)
Mean	0.245	0.141	0.141	0.209	0.223	1968.966	672.230	0.369	151.725	0.492	6.973
Obs	336	336	336	334	336	336	336	336	333	333	335
P(Male=Female)	0.875	0.364	0.364	0.720	0.434	0.979	0.539	0.601	0.988	0.498	0.367

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects on individual economic production, household wealth in 2012, empowerment and human capital of first generation females. Components of women's empowerment index are described in [Appendix B](#). Standard errors are clustered at the pre-program village level. All regressions include individual characteristics and pre-intervention characteristics interacted with birth cohort and are weighted to correct for attrition between birth and the 2012 MHSS2 survey. Individual characteristics include year of birth fixed effects and religion. Pre-intervention characteristics include all 1982-88 cohort individual and household characteristics listed in Table 1.

Appendix A Appendix Figures and Tables

FIGURE A1 — TRENDS IN CONTRACEPTIVE PREVALENCE RATE (CPR) AND MEASLES VACCINATION RATES (MVR)



Notes: This figure shows the Contraceptive Prevalence Rate (CPR) and Measles Vaccination Rate (MVR) for children aged 12-59 months by year. Contraceptive use data is from van Ginneken et al. (1998); Measles vaccination data from icddr,b Record Keeping System.

TABLE A1 — FIRST GENERATION: ITT EFFECTS ON COGNITION INDEX AND ITS COMPONENTS

	Pooled				Males				Females			
	Index (z-score)	MMSE (z-score)	Digit Spans (z-score)	Ravens (z-score)	Index (z-score)	MMSE (z-score)	Digit Spans (z-score)	Ravens (z-score)	Index (z-score)	MMSE (z-score)	Digit Spans (z-score)	Ravens (z-score)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Treat	-0.002 (0.041)	-0.007 (0.044)	0.008 (0.060)	-0.008 (0.052)	0.021 (0.058)	0.036 (0.071)	0.008 (0.079)	0.013 (0.083)	-0.015 (0.052)	-0.044 (0.057)	0.022 (0.075)	-0.022 (0.081)
Obs	1496	1494	1495	1496	692	691	691	692	804	803	804	804

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects on the components of the Cognition Index of the first generation. Standard errors are clustered at the treatment village level. All regressions include individual characteristics and pre-intervention characteristics interacted with birth cohort and are weighted to correct for attrition between birth and the 2012 MHSS2 survey. Individual characteristics include year of birth fixed effects, age cohort fixed effects and controls for religion. Pre-intervention characteristics include all individual and household characteristics in Table 1. Cognition sample sizes are smaller for men because no cognitive tests, with the exception of digit span forward, were administered in the phone survey. All females in the sample were interviewed in person.

TABLE A2 — ITT EFFECTS ON HUMAN CAPITAL OF FIRST GENERATION WITH HEALTH ENDOWMENT INFORMATION

	Height (cm) (1)	Short Stature (=1) (2)	Cognition Index (z-score) (3)	Grades Attained (years) (4)
<i>Panel A: Pooled</i>				
Treat	1.136 (0.357)**	-0.018 (0.018)	0.017 (0.044)	0.498 (0.224)*
Obs	1544	1544	1412	1549
<i>Panel B: Males</i>				
Treat	0.996 (0.542)+	-0.001 (0.022)	0.054 (0.059)	1.075 (0.334)**
Obs	783	783	652	786
<i>Panel C: Females</i>				
Treat	1.245 (0.433)**	-0.028 (0.028)	-0.013 (0.055)	-0.080 (0.215)
Obs	761	761	760	763

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows the results in Table 2 is robust to the exclusion of observations without health endowment information. Standard errors are clustered at the pre-program village level and reported in parentheses. All regressions include individual and pre-intervention characteristics interacted with birth cohort and are weighted to correct for attrition between birth and the 2012 MHSS2 survey. Individual characteristics include year of birth fixed effects, age cohort fixed effects, and religion. Pre-intervention characteristics include all characteristics in Table 1.

TABLE A3 — SECOND GENERATION: BALANCE OF MOTHERS' PRE-PROGRAM CHARACTERISTICS

	Treatment Area		Comparison Area		Difference		
	Mean	SD	Mean	SD	Mean	T-stat	Mean/SD
<i>Mother's Characteristics</i>							
Birth Year	1984.78	1.97	1984.91	1.76	-0.12	-0.84	-0.05
Islamic (=1)	0.84	0.62	0.95	0.26	-0.11	-3.03	-0.17
<i>Household Characteristics</i>							
Bari size	9.04	7.47	8.11	7.80	0.93	1.59	0.09
Family size	6.75	2.76	6.37	3.01	0.38	1.70	0.09
Asset Index (max=6)	2.79	1.59	2.75	1.71	0.05	0.38	0.02
Wall tin or tinmix (=1)	0.26	0.45	0.29	0.43	-0.03	-0.93	-0.05
Tin roof (=1)	0.80	0.47	0.82	0.46	-0.02	-0.58	-0.03
Number of rooms per capita	0.22	0.10	0.23	0.11	-0.01	-0.69	-0.04
Drinking water, tubewell (=1)	0.26	0.56	0.14	0.52	0.12	2.90	0.16
Drinking water, tank (=1)	0.43	0.84	0.33	0.96	0.10	1.42	0.08
HH age	48.39	15.67	45.29	15.71	3.10	2.56	0.14
HH <2 years of education	0.63	0.54	0.63	0.55	0.00	0.03	0.00
HH works in agriculture (=1)	0.57	0.46	0.55	0.59	0.02	0.54	0.03
HH works in fishing (=1)	0.08	0.37	0.09	0.33	-0.01	-0.55	-0.03
HH works in business (=1)	0.08	0.26	0.08	0.30	0.00	-0.20	-0.01
HH spouse's age	37.33	13.32	35.22	13.36	2.12	2.06	0.11
HH spouse <2 years of education	0.88	0.34	0.88	0.33	-0.01	-0.29	-0.02
1982 Land size	9.23	13.44	9.81	14.54	-0.58	-0.54	-0.03
Obs	307		372				

Notes: The sample includes first generation females who have a firstborn child in the second generation analytic sample. Unless otherwise noted, household characteristics come from the 1974 census. Standard deviations (SD) are clustered at the treatment village level. Household head and spouse age are reported, but these variables are likely affected by the family planning program increasing birth intervals and decreasing family size in the treatment area. Observations are weighted to correct for attrition between birth and the 2012 MHSS2 survey.

TABLE A4 — SECOND GENERATION: ITT EFFECTS ON INDEX COMPONENTS

	Index	Ages 0-6			Ages 7-14					
		Language	Fine Motor	Gross Motor	MMSE	Memory	Digits Forwards	Digits Backwards	Ravens	Matching
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
<i>Panel A: Pooled</i>										
Mother's Treat	0.001 (0.073)	0.052 (0.118)	0.003 (0.143)	-0.076 (0.126)	-0.139 (0.148)	-0.059 (0.101)	0.013 (0.097)	0.065 (0.141)	0.007 (0.154)	-0.146 (0.152)
Obs	596	353	348	342	266	478	488	267	265	266
<i>Panel B: Males</i>										
Mother's Treat	-0.100 (0.110)	0.109 (0.173)	-0.080 (0.197)	-0.241 (0.155)	-0.425 (0.275)	-0.017 (0.150)	-0.024 (0.152)	-0.252 (0.177)	-0.143 (0.215)	-0.509 (0.203)*
Obs	294	167	167	164	137	238	243	137	137	137
<i>Panel C: Females</i>										
Mother's Treat	0.166 (0.086)+	0.033 (0.154)	0.099 (0.177)	0.041 (0.186)	0.365 (0.200)+	-0.046 (0.172)	0.049 (0.125)	0.357 (0.213)+	0.332 (0.284)	0.302 (0.287)
Obs	302	186	181	178	129	240	245	130	128	129

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects on components of the cognition index of second generation children of first generation females using their mother's treatment assignment. Standard errors in parentheses are clustered at the mother's pre-program village level. All variables are internally standardized by sex and age (6 month bins for ages 0-6, 12 months for ages 7-14) using the comparison group's mean and standard deviation. Indexes are the average of the standardized components in subsequent columns. All models include six-month age fixed effects, mother's age fixed effects, mother's religion and mother's pre-intervention characteristics listed in Table A3.

TABLE A5 — FIRST GENERATION: ROBUSTNESS CHECKS

	Pooled				Males				Females			
	Height (cm)	Short Stature (=1)	Cognition Index (z-score)	Grades Attained (years)	Height (cm)	Short Stature (=1)	Cognition Index (z-score)	Grades Attained (years)	Height (cm)	Short Stature (=1)	Cognition Index (z-score)	Grades Attained (years)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
<i>Panel A: Base Results</i>												
Treat	1.008 (0.340)**	-0.018 (0.017)	-0.002 (0.041)	0.446 (0.216)*	0.961 (0.527)+	-0.008 (0.021)	0.021 (0.058)	0.969 (0.326)**	1.090 (0.416)**	-0.025 (0.026)	-0.015 (0.052)	-0.077 (0.218)
Obs	1633	1633	1496	1638	828	828	692	831	805	805	804	807
<i>Panel B: North Comparison Area</i>												
Treat	1.036 (0.403)*	-0.028 (0.022)	0.014 (0.051)	0.369 (0.265)	0.955 (0.566)+	-0.017 (0.027)	0.055 (0.063)	0.838 (0.380)*	1.054 (0.532)*	-0.033 (0.034)	-0.034 (0.069)	-0.076 (0.288)
Obs	1188	1188	1091	1194	603	603	507	607	585	585	584	587
<i>Panel C: West Comparison Area</i>												
Treat	0.865 (0.422)*	0.000 (0.018)	-0.017 (0.050)	0.493 (0.249)+	0.824 (0.680)	0.003 (0.023)	-0.007 (0.082)	1.045 (0.430)*	0.998 (0.488)*	-0.004 (0.027)	-0.001 (0.055)	-0.035 (0.231)
Obs	1180	1180	1081	1186	596	596	498	601	584	584	583	585
<i>Panel D: Add Extended Controls</i>												
Treat	0.882 (0.355)*	-0.013 (0.017)	-0.024 (0.043)	0.321 (0.200)	0.820 (0.517)	-0.004 (0.021)	-0.010 (0.064)	0.923 (0.322)**	1.015 (0.427)*	-0.024 (0.027)	-0.023 (0.053)	-0.208 (0.205)
Obs	1632	1632	1495	1637	827	827	691	830	805	805	804	807
	⋮											

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT estimates for the first generation are robust to a variety of potential threats. Panel A replicates the results of Table 2. Panels B and C restrict the sample of comparison units to those living in the North and West comparison areas, respectively (see Figure 1). Panel D adds controls for arsenic exposure above 100ppb, living in a village that was fully or partially eroded, access to the BRAC microcredit experiment, access to primary school and village access to healthcare [family welfare clinic, family welfare assistant, midwife, and alopatic] as of the individual's year of birth. Standard errors are clustered at the pre-program village level and reported in parentheses.

TABLE A5 — FIRST GENERATION: ROBUSTNESS CHECKS (CONT.)

	Pooled				Males				Females			
	Height (cm)	Short Stature (=1)	Cognition Index (z-score)	Grades Attained (years)	Height (cm)	Short Stature (=1)	Cognition Index (z-score)	Grades Attained (years)	Height (cm)	Short Stature (=1)	Cognition Index (z-score)	Grades Attained (years)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
⋮												
<i>Panel E: Unbalanced Controls</i>												
Treat	1.088 (0.412)**	-0.017 (0.017)	-0.011 (0.040)	0.366 (0.235)	0.969 (0.513)+	-0.007 (0.021)	0.009 (0.060)	0.845 (0.317)**	1.028 (0.388)**	-0.026 (0.025)	-0.010 (0.049)	-0.114 (0.249)
Obs	1633	1633	1496	1638	828	828	692	831	805	805	804	807
<i>Panel F: Only Muslims</i>												
Treat	1.062 (0.369)**	-0.020 (0.017)	0.005 (0.043)	0.461 (0.220)*	1.096 (0.559)+	-0.021 (0.021)	0.002 (0.061)	0.948 (0.326)**	1.037 (0.440)*	-0.016 (0.026)	0.008 (0.052)	-0.036 (0.226)
Obs	1484	1484	1357	1489	757	757	631	760	727	727	726	729
<i>Panel G: Double Difference</i>												
Treat × 1 (Born 1982-88)	0.985 (0.345)**	-0.042 (0.022)+	-0.013 (0.049)	0.396 (0.207)+	1.086 (0.621)+	-0.039 (0.029)	0.062 (0.072)	0.941 (0.378)*	1.016 (0.467)*	-0.042 (0.034)	-0.075 (0.067)	-0.109 (0.239)
Treat	0.017 (0.285)	0.021 (0.018)	0.019 (0.042)	0.038 (0.158)	-0.032 (0.422)	0.025 (0.022)	-0.024 (0.057)	-0.035 (0.262)	-0.025 (0.344)	0.018 (0.025)	0.063 (0.046)	0.089 (0.164)
Obs	4211	4211	4037	4340	1893	1893	1723	1952	2318	2318	2314	2388
<i>Panel H: Unweighted</i>												
Treat	0.981 (0.349)**	-0.018 (0.017)	0.001 (0.041)	0.450 (0.214)*	0.899 (0.538)+	-0.005 (0.021)	0.016 (0.058)	0.939 (0.324)**	1.089 (0.412)**	-0.028 (0.025)	-0.000 (0.051)	-0.036 (0.213)
Obs	1633	1633	1496	1638	828	828	692	831	805	805	804	807
<i>Panel I: Add Mother's Human Capital</i>												
Treat	1.344 (0.294)**	-0.027 (0.017)	0.006 (0.037)	0.422 (0.194)*	1.341 (0.485)**	-0.017 (0.020)	0.013 (0.055)	0.809 (0.301)**	1.369 (0.389)**	-0.034 (0.025)	0.010 (0.049)	0.028 (0.205)
Obs	1633	1633	1496	1638	828	828	692	831	805	805	804	807

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table is continued from the previous. Panel E includes year of birth fixed effects, cohort fixed effects and controls for religion but limits the pre-intervention characteristics to only those that were imbalanced (family size, tubewell access, household head individual and spouse characteristics [ages in 1974 and an indicators for less than two years of education]). Panel F estimates the original models using the subsample of Muslim respondents. Panel G estimates the ITT effect with a double difference model using people born 1947-1969 as an unaffected cohort. Panel H does not reweight for attrition between birth and the 2012 MHSS2 survey. Panel I includes controls for mother's height and education in 1996. We do not control for MMSE as it was not collected for all 1996 respondents but the results for the subsample including all three measures of human capital are qualitatively similar (available upon request). Standard errors are clustered at the pre-program village level and reported in parentheses.

TABLE A6 — FIRST GENERATION: KLING-LEIBMAN ATTRITION BOUNDING

	Height (cm)					Cognition Index (z-score)					Grades Attained (years)				
	Base	0.10 SD		0.25 SD		Base	0.10 SD		0.25 SD		Base	0.10 SD		0.25 SD	
		+	-	+	-		+	-	+	-		+	-		
		(1)	(2)	(3)	(4)		(5)	(6)	(7)	(8)		(9)	(10)	(11)	(12)
<i>Panel A: Pooled</i>															
Treat	1.008 (0.340)**	1.062 (0.309)**	0.843 (0.305)**	1.226 (0.313)**	0.679 (0.303)*	-0.010 (0.042)	0.008 (0.038)	-0.022 (0.039)	0.030 (0.038)	-0.045 (0.039)	0.446 (0.216)*	0.495 (0.200)*	0.369 (0.200)+	0.590 (0.200)**	0.274 (0.201)
Mean of Comparison	157.554	157.490	157.599	157.408	157.680						7.179	7.147	7.211	7.098	7.259
Obs	1633	1800	1800	1800	1800	1499	1662	1662	1662	1662	1638	1792	1792	1792	1792
<i>Panel B: Males</i>															
Treat	0.961 (0.527)+	1.056 (0.484)*	0.827 (0.481)+	1.228 (0.487)*	0.656 (0.480)	0.003 (0.059)	0.025 (0.053)	-0.007 (0.053)	0.050 (0.053)	-0.031 (0.054)	0.969 (0.326)**	1.014 (0.303)**	0.879 (0.302)**	1.116 (0.304)**	0.778 (0.302)*
Mean of Comparison	163.496	163.441	163.550	163.360	163.632						7.084	7.049	7.119	6.996	7.171
Obs	828	919	919	919	919	695	782	782	782	782	831	913	913	913	913
<i>Panel C: Females</i>															
Treat	1.090 (0.416)**	1.073 (0.364)**	0.872 (0.362)*	1.224 (0.367)**	0.721 (0.362)*	-0.015 (0.052)	-0.003 (0.048)	-0.030 (0.049)	0.017 (0.048)	-0.051 (0.049)	-0.077 (0.218)	-0.029 (0.200)	-0.139 (0.201)	0.054 (0.200)	-0.222 (0.202)
Mean of Comparison	151.397	151.343	151.451	151.261	151.533						7.276	7.247	7.305	7.203	7.349
Obs	805	881	881	881	881	804	880	880	880	880	807	879	879	879	879

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects across different attrition bounding schemes. Base models replicate the effects shown in Table 2 while subsequent columns impute attritors' human capital measures with the within-sample mean $\pm$ 0.10 and 0.25 standard deviations (SD). Treatment effect estimates for the 44-65 year-old comparison group are omitted for brevity. Age 24-30 means are for the comparison group. Standard errors are clustered at the pre-program village level and reported in parentheses.

TABLE A7 — SECOND GENERATION: ROBUSTNESS CHECKS

	Pooled			Males			Females		
	Height- for-Age (WHO z-score) (1)	Stunted (=1) (2)	Cognition Index (z-score) (3)	Height- for-Age (WHO z-score) (4)	Stunted (=1) (5)	Cognition Index (z-score) (6)	Height- for-Age (WHO z-score) (7)	Stunted (=1) (8)	Cognition Index (z-score) (9)
<i>Panel A: Base Results</i>									
Mother's Treat	0.106 (0.103)	-0.045 (0.035)	0.001 (0.073)	-0.034 (0.158)	0.002 (0.061)	-0.100 (0.110)	0.337 (0.147)*	-0.140 (0.048)**	0.166 (0.086)+
Obs	629	629	596	311	311	294	318	318	302
<i>Panel B: North Comparison Area</i>									
Mother's Treat	0.145 (0.120)	-0.053 (0.043)	0.014 (0.084)	0.078 (0.213)	-0.070 (0.083)	-0.028 (0.121)	0.377 (0.185)*	-0.108 (0.072)	0.112 (0.118)
Obs	455	455	433	235	235	225	220	220	208
<i>Panel C: West Comparison Area</i>									
Mother's Treat	0.113 (0.125)	-0.040 (0.039)	-0.028 (0.088)	-0.098 (0.145)	0.068 (0.060)	-0.242 (0.140)+	0.430 (0.189)*	-0.194 (0.059)**	0.209 (0.114)+
Obs	466	466	438	222	222	208	244	244	230
<i>Panel D: Add Extended Controls</i>									
Mother's Treat	0.122 (0.110)	-0.045 (0.038)	-0.019 (0.073)	-0.000 (0.172)	-0.009 (0.072)	-0.132 (0.115)	0.335 (0.147)*	-0.131 (0.052)*	0.159 (0.088)+
Obs	626	626	593	309	309	292	317	317	301
<i>Panel E: Mother's Unbalanced Controls</i>									
Mother's Treat	0.090 (0.102)	-0.037 (0.035)	0.006 (0.066)	-0.026 (0.159)	0.006 (0.058)	-0.128 (0.102)	0.288 (0.145)*	-0.115 (0.046)*	0.212 (0.070)**
Obs	629	629	596	311	311	294	318	318	302
<i>Panel F: Only Muslims</i>									
Mother's Treat	0.016 (0.105)	-0.013 (0.032)	-0.007 (0.077)	-0.221 (0.178)	0.082 (0.060)	-0.162 (0.126)	0.278 (0.152)+	-0.137 (0.049)**	0.171 (0.093)+
Obs	573	573	544	276	276	262	297	297	282
<i>Panel G: Double Difference</i>									
Mother's Treat $\times \mathbb{1}$ (Mother Born 1982-88)	0.075 (0.113)	-0.031 (0.043)	0.090 (0.118)	-0.148 (0.149)	0.065 (0.064)	0.013 (0.151)	0.391 (0.161)*	-0.171 (0.061)**	0.243 (0.136)+
Mother's Treat	-0.011 (0.065)	-0.003 (0.027)	-0.102 (0.085)	0.026 (0.078)	-0.036 (0.034)	-0.123 (0.093)	-0.068 (0.084)	0.040 (0.036)	-0.113 (0.101)
Obs	2796	2796	1378	1433	1433	697	1363	1363	681
<i>Panel H: Unweighted</i>									
Mother's Treat	0.103 (0.102)	-0.043 (0.033)	-0.006 (0.073)	-0.038 (0.149)	0.005 (0.058)	-0.123 (0.110)	0.316 (0.150)*	-0.133 (0.047)**	0.166 (0.085)+
Obs	629	629	596	311	311	294	318	318	302
<i>Panel I: Control for MNCH Eligibility</i>									
Mother's Treat	0.027 (0.101)	-0.021 (0.039)	0.046 (0.075)	-0.184 (0.161)	0.076 (0.068)	-0.076 (0.119)	0.280 (0.164)+	-0.131 (0.058)*	0.203 (0.093)*
Obs	629	629	596	311	311	294	318	318	302

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects for the second generation are robust to a variety of potential threats. Panel A replicates the results of Table 3. Panels B and C restrict the sample of comparison units to those living in the North and West comparison areas, respectively (see Figure 1). Panel D includes controls for an index of available healthcare supply in the child's year of birth (family welfare clinic, family welfare assistant, midwife and allopathic), minimum distance to primary education (government, private and informal primary school), and distance to government secondary school. Panel E includes six month age fixed effects but limits the mother's attributes and preintervention characteristics to only those that were imbalanced (age, religion, family size, tubewell access, household head's age and household head's spouse's age). Panel F limits the sample children of Muslim mothers. Panel G estimates ITT effects with a double-difference specification on human capital outcomes of the second generation using similarly-aged children of mothers born prior to 1982 as the unaffected group. Panel H does not reweight for mother's attrition. Standard errors in parentheses are clustered at the pre-program village level. Panel I controls for mother's eligibility for MNCH based on her 2005 residence interacted with a dummy indicating if the child was born on or after 2007.

TABLE A8 — SECOND GENERATION: KLING-LEIBMAN AND WORST CASE ATTRITION BOUNDING

	Height-for-Age (z-score)					Stunting (=1)			Cognition Index (z-score)				
	Base	0.10 SD		0.25 SD		Base	Worst case		Base	0.10 SD		0.25 SD	
		+	-	+	-		+	-		+	-	+	-
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
<i>Panel A: Pooled</i>													
Mother's Treat	0.106 (0.103)	0.130 (0.097)	0.100 (0.096)	0.153 (0.097)	0.078 (0.096)	-0.045 (0.035)	0.013 (0.034)	-0.119** (0.034)	0.006 (0.073)	0.005 (0.067)	-0.013 (0.067)	0.020 (0.067)	-0.027 (0.066)
Observations	629	672	672	672	672	629	672	672	593	636	636	636	636
<i>Panel B: Males</i>													
Mother's Treat	-0.034 (0.158)	-0.016 (0.145)	-0.050 (0.145)	0.009 (0.146)	-0.076 (0.145)	0.002 (0.061)	0.072 (0.057)	-0.083 (0.060)	-0.095 (0.111)	-0.100 (0.102)	-0.123 (0.102)	-0.083 (0.102)	-0.140 (0.101)
Observations	311	339	339	339	339	311	339	339	293	321	321	321	321
<i>Panel C: Females</i>													
Mother's Treat	0.337* (0.147)	0.345* (0.139)	0.317* (0.139)	0.366** (0.139)	0.296* (0.140)	-0.140** (0.048)	-0.077 (0.048)	-0.194** (0.050)	0.171* (0.086)	0.160+ (0.082)	0.143+ (0.082)	0.172* (0.082)	0.131 (0.082)
Observations	318	333	333	333	333	318	333	333	300	315	315	315	315

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects across different attrition bounding schemes. Base models replicate the effects shown in Table 3 while subsequent columns impute attritors' human capital measures with the within-sample mean $\pm$ 0.10 and 0.25 standard deviations (SD). For the binary outcome stunting, we impute the attritors with ones (+) or zeros (-). Standard errors are clustered at the pre-program village level and reported in parentheses.

TABLE A9 — FIRST GENERATION: ITT EFFECTS ON COMPONENTS OF EMPOWERMENT INDEX FOR FEMALES

	Gender Equivalence (1=agree/0=disagree)					
	Practice purdah?	Girls equally educated?	Women allowed to visit relatives?	Women can divorce?	Women should support parents?	Husband & wife choose kids?
	(1)	(2)	(3)	(4)	(5)	(6)
Treat	-0.016 (0.010)	-0.007 (0.005)	-0.011 (0.061)	0.025 (0.050)	-0.011 (0.005)*	-0.010 (0.007)
Treat × Male Firstborn	0.004 (0.008)	0.012 (0.009)	0.005 (0.083)	-0.068 (0.067)	0.005 (0.012)	0.010 (0.007)
Male Firstborn (=1)	0.007 (0.005)	-0.005 (0.005)	0.023 (0.063)	0.052 (0.049)	0.001 (0.010)	-0.010 (0.007)
Mean of Comparison	1.000	0.003	0.566	0.661	0.011	0.005
Obs	673	673	673	673	673	673
	Women's Mobility (1=yes/0=no)					
	Visit other baris?	Vist other villages?	Use public transit?	Make small purchases?	Make large purchases?	
	(1)	(2)	(3)	(4)	(5)	
Treat	0.018 (0.028)	0.000 (0.021)	0.031 (0.026)	-0.043 (0.050)	0.074 (0.038)+	
Treat × Male Firstborn	-0.032 (0.040)	-0.046 (0.028)	-0.007 (0.030)	0.019 (0.077)	-0.023 (0.055)	
Male Firstborn (=1)	-0.000 (0.026)	0.014 (0.020)	0.025 (0.023)	0.012 (0.050)	-0.001 (0.041)	
Mean of Comparison	0.916	0.965	0.954	0.423	0.822	
Obs	676	676	675	675	676	
	Husband justified in physically harming wife if she... (1=yes/0=no)					
	Goes out w/o telling him?	Doesn't obey elders?	Neglects her children?	Refuses him sex?		
	(1)	(2)	(3)	(4)		
Treat	-0.011 (0.034)	-0.013 (0.036)	0.005 (0.035)	-0.010 (0.017)		
Treat × Male Firstborn	-0.004 (0.051)	-0.013 (0.049)	-0.012 (0.048)	0.011 (0.025)		
Male Firstborn (=1)	-0.007 (0.035)	0.022 (0.037)	0.019 (0.029)	0.009 (0.017)		
Mean of Comparison	0.114	0.152	0.106	0.027		
Obs	673	673	673	673		
	Do you have the final say about... (1=yes/0=no)					
	Major purchases?	Spouse's healthcare?	Own healthcare?	Visiting relatives?		
	(1)	(2)	(3)	(4)		
Treat	-0.023 (0.053)	-0.001 (0.052)	-0.014 (0.039)	0.033 (0.045)		
Treat × Male Firstborn	0.087 (0.065)	-0.066 (0.063)	0.027 (0.052)	0.017 (0.065)		
Male Firstborn (=1)	-0.082 (0.046)+	0.045 (0.046)	0.001 (0.034)	-0.037 (0.047)		
Mean of Comparison	0.303	0.292	0.176	0.211		
Obs	675	675	675	675		

Notes: + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$. This table shows ITT effects for the components of the women's empowerment index for the sample of first generation females with at least one live birth. *Male Firstborn* is an indicator for whether the female's firstborn child is male. All regressions include mother's individual and pre-intervention characteristics and are weighted to correct for attrition between birth and the 2012 MHSS2 survey. Standard errors are clustered at the pre-program village level and reported in parentheses.

Appendix B Data and Construction of Selected Variables

This appendix describes the data sources, attrition, and the creation of main outcome variables.

B.1 Data Sources

MHSS1/2.—The main outcomes variables used in this paper are drawn from MHSS2. MHSS2 is a large socioeconomic survey comprised of several instruments including a household survey, a village survey, facility surveys, and market price survey. The household survey contains two modules that is given to individuals: one, the individual module, collected information on individuals in a question format, and the another, the testing module, collected information on individuals from tests including the anthropometric and cognitive tests used in this paper. Educational outcomes were collected in the individual module of the household survey, as well as, by proxy in the household roster. Most of the data were collected during face-to-face interviews in the homes of the respondents, though a subset of data was collected in a phone survey of international migrants who did not return to Bangladesh during the data collection period (about 15 percent of our male sample). The MHSS2 phone survey instrument was shorter than the in-person survey instrument, but includes grades of education attained, self-report height and weight, and digit span forward. The remaining measures of cognition examined in this paper are not in the phone survey so sample sizes are smaller for these outcomes.

MHSS2 was conducted between 2012 and 2014 and was designed to be a panel to MHSS1 (icddr,b 1996). MHSS1 is a seven percent random subsample of household compounds (called *baris*) from the study area and was designed to be representative of the study area's 1996 population. In MHSS1, two households were interviewed in each *bari*: a primary household, selected randomly, and a secondary household, selected purposively. Within a household, individuals aged six and older were randomly selected to be personally interviewed.

The MHSS2 sample includes all individuals selected for personal interview in MHSS1 primary households creating panel data for these individuals.²⁷ To limit migration selection for key age groups, the MHSS2 sample included individuals born between 1972 and 1989 to a MHSS1 primary household that had migrated out of Matlab between 1977 and 1996 (referred to as pre-1996 migrants).²⁸ To the extent that a whole household and lineage migrated out of Matlab between the start of the program and 1996, leaving no one in that lineage available for selection into the MHSS1 sample, the MHSS2 sample could still suffer from migration selection. The annual migration rate of whole households from the study areas prior to 1996 was low, 0.66 percent, and most of the exiting households were Hindu.

With the exception of the phone survey, all tests were collected on the individuals in their home by well-trained testers. Testers were extensively trained to implement protocols in a similar fashion. Retraining took place a couple of times throughout the survey period to restandardize the testers. The testers were generally female, though there were a few male testers who implemented the tests on adult male respondents. For the most part, testers were only responsible for implementing the test module and collected information a few weeks after the household survey. There were a few enumerators who were trained on the entire household survey, including the testing module, to interview adult migrants who were living on their own.

Census Data.—Periodic censuses were collected for all individuals in the study area (treatment and comparison areas) by icddr,b. These data typically include household location, household characteristics and composition, employment, education, and assets. We obtain pre-program individual and household

27. MHSS2 sample also included all panel member descendants, and their co-resident spouses. However, spouses of panel members who had migrated out of the study areas were not tracked for interview.

28. The pre-1996 migrants were identified by using the detailed demographic surveillance data.

data on the analysis sample from the 1974 census (icddr,b 1974) and use these data to test for differences in baseline characteristics between the treatment and comparison areas. We also use the 1974 and 1982 censuses (icddr,b 1982) to link individuals to the study area before 1977 to construct an individual's intent-to-treat status (see section C below).

Demographic Surveillance Site (DSS) Data.—Vital registration data provide prospective tracking of every birth, death, marriage, divorce, and in- and out- migration occurring in the study area. As such, we know when someone enters and leaves the study area. Data were collected by icddr,b and are high quality in part because they were collected so frequently: every two weeks until 1997, every month between 1998 and 2006, and every two months between 2007 and MHSS2. These data include pre-program data from 1974 onwards and are used to construct birth dates and an individual's intent-to-treat status. In addition, we use these data to construct pre-program migration network variables for each individual in the analysis sample, as well as, out-migration variables such as whether someone has ever migrated, and out-migration variables for years not covered in the MHSS2 migration history.

B.2 Attrition

Table B1 and B2 document attrition rates for the first and second generations. Relative to other long-term follow-up surveys, attrition in the MHSS2 sample is low. For the first generation, including death and any other type of non-response, the attrition rate at the household level is 7 percent. Attrition rates are slightly higher for variables from the testing module at 10 percent for men and 7 percent for females for height information (Table B1). Attrition rates for men reached 24 percent for outcomes that were not collected in the phone survey. Differences in attrition rates between the treatment and comparison areas for the full sample are less than one percent and are not statistically different from each other. For the second generation, attrition is similarly low. Of the 677 females aged 24-30 with a firstborn child, 629 were surveyed and had a valid height measurement. Only 47 children were lost (6.92% attrition rate), 4 of which had invalid height measurements, 29 died before being surveyed and 14 were born to migrant females whose children were not followed. The attrition rate did not differ between treatment and comparison areas by any definition or among males and females separately.

Even with low levels of attrition and the rate balanced between treatment and comparison areas, we would be concerned if there was differential attrition between treatment and comparison groups as this selection could bias the results. Table B3 shows individual and household characteristics do not predict attrition status nor do their interactions with treatment status with a few exceptions: drinking water source and whether the household head works in business are associated with high levels of treatment area attrition. The differences are not statistically significant when we split the sample by sex and we can comfortably fail to reject the null hypothesis that all interactions between treatment status and baseline characteristics are jointly zero for the pooled and male sample. For females, selection into attrition was slightly negative based on human capital: the household head was 7 percentage points more likely to have less than two years of education. We conclude there is limited differential attrition between the treatment and comparison areas but include the attributes attrition weight predictions to account for any possible selection.

The low attrition rates are a result of a carefully designed tracking protocol. Migrants were tracked all over Bangladesh, and a rapid response system was developed that allowed trackers in Matlab to connect enumerators placed in different parts of the country with respondents who had left Matlab. Intensive interviewing took place during all the Eid holidays from 2012–2014. Survey teams targeted international migrants, far away domestic migrants, and hard-to-track migrants returning to Matlab for the holiday. Finally, a phone survey was employed to collect information on a subset of questions from the main survey from male international migrants who did not return to Bangladesh during the survey period.

While there is a limited set of variables available for this group in the phone survey, most educational and anthropometric outcomes, including self-report height and weight, used in this study were collected during the phone survey. However, with the exception of digit span forward, no other cognition or health measures used in the study were included in the phone survey.

TABLE B1 — FIRST GENERATION: 2012 ATTRITION RATES OF 1982-1988 COHORT

	Pooled			Males			Females		
	Attrition Rate	T-C Difference		Attrition Rate	T-C Difference		Attrition Rate	T-C Difference	
		Mean	(SE)		Mean	(SE)		Mean	(SE)
Not found or refused	5.61%	-0.007	(0.013)	5.11%	0.008	(0.018)	6.13%	-0.023	(0.016)
Not found, refused, or dead	7.22%	-0.008	(0.014)	6.96%	0.005	(0.019)	7.49%	-0.020	(0.017)
Non-missing height information	9.44%	-0.003	(0.015)	10.01%	0.011	(0.020)	8.85%	-0.017	(0.019)
Non-missing MMSE information	17.00%	-0.012	(0.017)	24.59%	-0.013	(0.029)	9.08%	-0.012	(0.019)
Non-missing height, no phone survey	16.78%	-0.006	(0.017)	24.37%	0.004	(0.028)	8.85%	-0.017	(0.019)

Notes: Sample includes 1982-1988 cohort. The Attrition Rate column displays the percent of the sample who attrited, while mean and SE show the difference and standard error of the difference in attrition rates by treatment and comparison area. The standard error of the difference is clustered at the pre-program village level. There are 919 men and 881 females across the two cohorts in the sample frame. Missing height information indicates the respondent was not able to be measured, including not found, refused, or dead, but outliers are not dropped here as they are in the final analysis.

TABLE B2 — SECOND GENERATION: 2012 ATTRITION RATES

	Pooled				Males				Females			
	Number Attrited	Attrition Rate	T-C Difference		Number Attrited	Attrition Rate	T-C Difference		Number Attrited	Attrition Rate	T-C Difference	
			Mean	(SE)			Mean	(SE)			Mean	(SE)
Child missing height	47	6.92%	-0.037	(0.020)	29	8.48%	-0.060	(0.030)	18	5.34%	-0.014	(0.025)
Child not found	43	6.33%	-0.032	(0.019)	28	8.19%	-0.054	(0.030)	15	4.45%	-0.010	(0.023)
Child died	29	4.27%	-0.024	(0.016)	22	6.43%	-0.034	(0.027)	7	2.08%	-0.014	(0.016)
Mother migrated	14	2.06%	-0.008	(0.011)	6	1.75%	-0.020	(0.014)	8	2.37%	0.004	(0.017)

Notes: Sample includes live firstborn children of 1982-1988 females. Number Attrited is the number of observations in the sample who attrited, while Attrition Rate displays the percent of the sample who attrited, and mean and SE show the difference and standard error of the difference in attrition rates by treatment and comparison area. The standard error of the difference is clustered at the pre-program village level. There are 342 males and 337 females in the sample frame. Missing height information indicates the respondent was not able to be measured, including not found, refused, or dead, but outliers are not dropped here as they are in the final analysis.

TABLE B3 — FIRST GENERATION: ATTRITION BALANCE OF 1982-1988 COHORT'S INDIVIDUAL AND PRE-PROGRAM CHARACTERISTICS BY TREATMENT STATUS

	Pooled		Males		Females	
	Main Effect	Interaction	Main Effect	Interaction	Main Effect	Interaction
Male (=1)	0.007 (0.020)	0.025 (0.026)				
Birth year	-0.002 (0.005)	-0.003 (0.008)	-0.002 (0.008)	-0.003 (0.011)	-0.000 (0.008)	-0.003 (0.010)
Islamic (=1)	0.006 (0.055)	0.027 (0.065)	0.107 (0.052)*	-0.116 (0.079)	-0.076 (0.086)	0.140 (0.091)
HH Bari size	0.002 (0.001)	0.001 (0.002)	0.001 (0.002)	0.001 (0.004)	0.003 (0.003)	0.002 (0.004)
HH Family size	0.007 (0.005)	-0.005 (0.007)	0.004 (0.007)	0.002 (0.010)	0.009 (0.008)	-0.011 (0.010)
Latrine (=1)	-0.062 (0.037)+	0.057 (0.050)	-0.043 (0.047)	0.033 (0.066)	-0.069 (0.054)	0.068 (0.064)
Owns a lamp (=1)	-0.008 (0.020)	0.015 (0.036)	-0.008 (0.035)	0.002 (0.059)	-0.004 (0.031)	0.011 (0.037)
Owns a watch (=1)	-0.011 (0.025)	0.035 (0.043)	-0.015 (0.033)	0.026 (0.058)	0.004 (0.036)	0.030 (0.058)
Owns a radio (=1)	-0.058 (0.033)+	0.035 (0.052)	-0.001 (0.050)	0.021 (0.081)	-0.108 (0.040)**	0.054 (0.062)
Number of cows	0.002 (0.007)	0.000 (0.012)	-0.006 (0.009)	0.005 (0.015)	0.008 (0.010)	-0.004 (0.015)
Number of boats	-0.029 (0.022)	0.018 (0.029)	-0.013 (0.030)	-0.028 (0.046)	-0.052 (0.034)	0.072 (0.042)+
Wall tin or tinmix (=1)	-0.009 (0.025)	0.032 (0.033)	-0.011 (0.038)	0.026 (0.054)	-0.003 (0.034)	0.048 (0.049)
Tin roof (=1)	-0.029 (0.027)	0.009 (0.042)	-0.020 (0.040)	0.002 (0.066)	-0.042 (0.045)	0.010 (0.059)
Number of rooms per capita	-0.008 (0.123)	-0.082 (0.177)	-0.011 (0.159)	0.001 (0.220)	-0.005 (0.158)	-0.194 (0.261)
Drinking water, tubewell (=1)	-0.016 (0.028)	0.071 (0.041)+	-0.027 (0.037)	0.092 (0.059)	-0.008 (0.038)	0.082 (0.055)
Drinking water, tank (=1)	-0.017 (0.022)	0.074 (0.037)*	-0.022 (0.026)	0.079 (0.046)+	-0.008 (0.034)	0.076 (0.057)
HH age	-0.002 (0.001)*	0.002 (0.002)	-0.003 (0.001)*	0.002 (0.002)	-0.002 (0.002)	0.003 (0.002)
HH <2 years of education (=1)	-0.005 (0.021)	0.040 (0.033)	0.009 (0.027)	0.005 (0.043)	-0.024 (0.032)	0.090 (0.043)*
HH works in agriculture (=1)	0.017 (0.025)	0.005 (0.033)	-0.001 (0.031)	0.043 (0.051)	0.037 (0.038)	-0.031 (0.049)
HH works in fishing (=1)	0.111 (0.063)+	-0.052 (0.105)	0.134 (0.094)	-0.176 (0.128)	0.117 (0.076)	0.014 (0.117)
HH works in business (=1)	-0.043 (0.032)	0.079 (0.042)+	-0.055 (0.049)	0.106 (0.069)	-0.037 (0.045)	0.035 (0.073)
HH spouse's age	0.003 (0.001)*	-0.003 (0.002)	0.002 (0.002)	-0.001 (0.003)	0.004 (0.002)+	-0.007 (0.003)*
HH spouse <2 years of education (=1)	-0.002 (0.029)	0.001 (0.041)	-0.026 (0.048)	0.028 (0.065)	0.020 (0.039)	-0.027 (0.059)
1982 Land size	0.001 (0.001)	-0.001 (0.001)	0.001 (0.001)	-0.003 (0.001)*	0.000 (0.001)	0.002 (0.002)
F-stat all interactions = 0		1.27		1.29		1.78
P-value		0.20		0.19		0.02
Obs		1800		919		881

Notes: This table shows the results of regression attrition status on a fully-interacted model of treatment status (main term not shown) and the listed individual and pre-program characteristics. Attrition is defined as missing height information (either measured or self-reported). The *F*-statistic and associated *P*-value reported tests the null hypothesis that all interactions between characteristics and treatment status are jointly zero.

B.3 Construction of Selected Variables

B.3.1 First Generation

Height & Short Stature.—Adult respondents were measured standing using a Shorr height board. If the respondent was interviewed by phone, their height was self-reported. We dropped height values above 195cm and 182cm for males and females, respectively. We defined an individual's short stature if their height was less than 155cm for males or 145cm for females.

Cognition Index.—This index was created by taking the average of standardized MMSE, Digit Spans Forward and Backward, and Ravens scores. All three components were internally standardized using the comparison group mean and standard deviation by sex.

Empowerment Index.—We construct an index measuring the mother's relative empowerment from four subindices: decision making, gender equivalence in social issues, attitudes towards husband violence, and mobility. All measures are scored or recoded so that a larger value represents more empowerment of the woman. Each subindex and its components are described below.

Decision Making.—Married or previously married women were asked who had the final say in four categories: major household purchases, decisions about spousal health and treatment, decisions about their own health and treatment, and visits to family or relatives. We coded each response with 1 if the woman made decisions independently or jointly about the matter and 0 if someone else had the final say. Finally, we created an index by averaging the four responses.

Gender Equivalence in Social Issues.—Women were asked to give an individual opinion on six claims: (1) women should practice purdah (female seclusion), (2) girls should be educated as much as boys, (3) women should be allowed to go out alone, (4) women should have the right to initiate divorce, (5) daughters should support elderly parents and (6) husband and wife should jointly agree to have children. We coded responses to (1) as 0=agree, 0.5=partially agree and 1=do not agree while responses to (2)-(6) were coded in reverse so that a larger value represents a higher opinion of gender equivalence. We then took the average of all six responses to form an index of gender equivalence in social issues.

Attitudes Toward Husband Violence.—Women were asked whether a husband is justified in hurting his wife if he is ever angered or annoyed by his wife's behavior in four situations: (1) going out without telling her him, (2) disobeying family elders, (3) neglecting their children and (4) refusing to have sex. We recoded responses so that yes=0 and no=1 to represent a more egalitarian viewpoint and averaged her four responses to create an index of attitudes toward husband violence.

Women's Mobility.—Women were asked if in the past twelve months they had gone out alone or with someone else for five activities: (1) visiting women in other bars, (2) visiting someone outside their village, (3) visiting any place on public transport, (4) visiting the local store for small purchases (oil, rice, salt, etc.) and (5) visiting a store or market to make larger purchases such as clothing. We coded each response as 1 if they went alone, 0.5 if they went out with someone and 0 if they never went out. Then we averaged responses to create an index that represents more mobility.

B.3.2 Second Generation

Height-for-age and Stunting.—Second generation children were measured using the Shorr height board lying down if the child was less than two years old or shorter than 83cm and standing up if older than two or taller than 83cm. We calculated standardized height-for-age using WHO growth standards. Any

values more than five standard deviations from the mean were set to missing. Stunting was defined as a height-for-age of two standard deviations or more below the mean.²⁹

Cognition Index.—For children aged 0-6, we construct a Cognition Index by taking the average of standardized Denver Language, Fine Motor and Gross Motor skills. For children aged 7-14, we take the average of internally standardized MMSE, Memory score, Digit Spans Forward and Backward, Ravens, and Matching score. All components were standardized using the comparison group mean and standard deviation by sex and age in years.

Vaccinations.—BCG, DPT/Penta, Polio and MMR vaccine histories for children were recorded by questioning the mother or caretaker. We calculate the number of vaccination types (out of four) the child received.

Observed Positive Parenting.—For all ages, the enumerator observed how the parent or caretaker treated the child at hand. Eight binary responses were recorded to questions such as “Caretaker uses some term of endearment [...] for [child’s name]...” and “Did caretaker hit child?”. All responses were recoded so that an affirmative represented a qualitatively positive outcome. Scores were summed for a possible maximum of 8 and internally standardized.

Family Care and Home Observation Indicators.—A number of measures were recorded to characterize a child’s home life and interactions with their parent(s). We aggregated these measures into four groups and internally standardized the resultant indices using the comparison group’s mean and standard deviation within sex and six-month age bins to summarize a particular type of interaction. Each index is described below.

Parent Plays with Child.—For children aged 0-6, the caretaker was asked how many days out of the past seven they spent participating in the following five activities with the child: read books together; told stories, nursery rhymes, or poems; sang songs; played with toys; and spent time naming, counting, and/or drawing things. The number of days was aggregated across activities and standardized as described above.

Child Engages in Positive Play.—For children aged 0-6, the caretaker was asked what items the child played with. Four binary responses were recorded indicating if the child had played with each of the following in the past 30 days: toys that make or play music, materials for drawing and writing, role-playing as someone else (e.g. Mommy, doctor, teacher, etc) using toys, or toys that encouraged movement (e.g. ball, skipping rope, etc). Responses were summed (for a possible maximum of 4) and internally standardized.

Child-Related Books & Magazines.—For children aged 0-6, the number of books meant for children (not including school books) and the number of magazines and newspapers available in the home was recorded, summed, and internally standardized.

Appendix C Potential Confounders

We accounted for several other important and well-documented changes that occurred in Matlab over the 35-year period since program inception that could confound results.

29. Guideline: Assessing and Managing Children at Primary Health-Care Facilities to Prevent Overweight and Obesity in the Context of the Double Burden of Malnutrition: Updates for the Integrated Management of Childhood Illness (IMCI). Geneva: World Health Organization; 2017. Table 1, World Health Organization (WHO) classification of nutritional status of infants and children. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK487900/table/fm.s1.t1/>

The Bangladesh Female Secondary Education Stipend Program.—One government program that is pertinent for MCH-FP effects on education, and difficult to control for, is the Bangladesh Female Secondary Education Stipend Program. This was a national program that became available in Matlab in 1984 for females attending grades 6-10 who were unmarried, had 75 percent attendance and scored 45 percent on school exams. The program targeted individual females, not schools, and provided a stipend and covered many school costs. This program was available to all the females in the first generation sample in this paper for the entirety of their secondary schooling in both the treatment and comparison areas, providing no variation to test for heterogeneity treatment effects of MCH-FP based on differential access to this program. In addition, we do not have data on who received the stipend program, nor sufficient data to determine who may have been eligible. Shamsuddin (2015) estimates that five years of exposure to the program led to one year gain in education. As a result, any program effects on educational outcomes of the MCH-FP program on females needs to outweigh the already large effects on education from the female stipend program.

Erosion and Flood Risk.— In 1987 the government of Bangladesh completed the Meghna Dhonnogoda Irrigation Project. This project involved constructing a river embankment along the northern bank of the Meghna River where it meets the west bank of the smaller Dhonnogoda River, which runs through Matlab (see Figure 1). The villages near this project were all located in the comparison area, and the embankment had two important consequences for these villages. First, seven villages in this area lining the river were partially or fully inundated between 1984 and 1986 as part of the project. All households in these villages were displaced, with most initially relocating to adjoining villages within the comparison area. To control for potential differences in the Meghna area in general, we include a variable indicating if the person's treatment village was submerged as a result of the project.

1993 BRAC Microcredit Experiment.—In 1993, BRAC introduced an experiment that provided landless females with access to microcredit. The rollout was designed to include villages in both treatment and comparison areas, but the presence of the program could still bias our results. We include an indicator for whether the village ever participated in BRAC during its experimental period.

School Access.—There was a large expansion of education during the 1990s, including construction of primary schools. If education supply was not equal between the two areas, it could bias our results. We control for differential primary school access by including an indicator variable that is one if a primary school was present in the first generation individual's treatment village during the years that they were age-eligible. For the second generation analysis, we control for the minimum distance to nearest government, private and informal primary school and the distance to the nearest government secondary school.

Health Facility Access.—Access to health care supply could have changed over time differentially between the two areas. To control for post-program access to health care, for the first generation, we use information on access to health care from the MHSS1 village survey (taken in 1996). MHSS1 surveyed village leaders about health facilities used by people from their village including the date of construction. We construct indicators for the presence of a Family Welfare Center (a government clinic, FWC), a Family Welfare Assistant (a government health worker that travels to villages, FWA), a non-MBSS allopathic doctor, and a Trained Traditional Birth Attendant (a midwife) in the individual's birth year to account for differential access to pre- and post-natal care. We use the MHSS1 Village Survey rather than its MHSS2 equivalent to avoid selection from the timing of facility opening and closings. For the second generation, we use a similar questionnaire in MHSS2 to construct indicators for the same facility access in the year a second generation child was born and aggregate these indicators into a single index ranging 0-4.

Arsenic Exposure.—To account for differential access in tubewell water, which contains arsenic, between the treatment and comparison areas, we control for arsenic exposure in tube well water. We use 2003 measures of arsenic collected by icddr,b . We use the 2003 measure rather than measures collected at later times, including with MHSS2, because it was unknown that there were arsenic in the wells prior to 2003, so this represents a period before families engaged in well switching which could be correlated with treatment status, and since it was measured at a time closer to when the sample of interest were young children. Wells are linked to MHSS1 households using the ID of the person who takes care of the well. For household who do not take care of a well, we take the average arsenic level in the three closest wells. For households that reported not using a tubewell in MHSS1 (which was prior to knowing about arsenic in the well), the value of arsenic is set to zero. The included control is an indicator of an arsenic level above 100 parts per billion (ppb, micrograms per liter). Note a majority of the children in the sample were born after the wells were established, so the age fixed-effects control for the length of time exposed to the well water.

2001-2003 Maternal and Infant Nutritional Intervention (MINIMat).—Between November 2001 and October 2003 pregnant females in the study area were randomized into a nutritional intervention program. The randomization was done at the individual level and was conducted in both the MCH-FP treatment and comparison areas so should be balanced across areas and not drive results (El Arifeen et al., 2018). The children impacted by the MINIMat intervention were largely born before the second generation cohort that is considered in this study so should not impact results. We do drop any children born prior to 2003, and results remain the same.

2007 Maternal, Neonatal and Child Health Project (MNCH).— Beginning in 2007 this program aimed to increase facility-based childbirth and neonatal health. Pregnant women enrolled in the program were visited by a community health worker at 12-14 weeks and again between weeks 32 - 34. Women were encouraged to deliver at a facility with skilled obstetric care rather than at home. Postnatal visits were made by a community health worker on days 3, 7, and 28 to provide counseling on newborn and maternal well-being. See Rahman et al. (2011) for a detailed description of the program. The targeting of this program overlapped with the treatment area of the MCH-FP program, however children born before 2007 were born after the program so did not benefit. As such, the program would have mainly affected children less than age 6. For robustness, we control for mother's exposure to the MNCH program based on her residence in 2005 interacted with a dummy indicating whether the child was born on or after 2007. Results are similar when including this control and available upon request.

Appendix D Weights

The main results are weighted for attrition between birth and MHSS2 using inverse propensity weights. The analysis sample includes respondents from MHSS1 and individuals from MHSS1 households that had migrated out of the DSS area prior to the survey conducted in 1996. The main reason for non-response are migration in early adulthood and death primarily during infancy. Weights are constructed in two steps. First, we estimate weights to account for selection into the MHSS1 sample frame between birth and MHSS1, which is mainly a result of mortality. Second, we estimate weights to account for attrition of MHSS1 respondents in the MHSS2 survey. We estimate these two probabilities separately and then multiple them to obtain a weight to account for attrition between birth and MHSS2.

The weight the account for attrition between birth and MHSS1 estimate the conditional probability that an individual born in the study site was present to be surveyed in MHSS1 using demographic surveillance data. To do this, we assign treatment status to the universe of individuals born in the study

site between 1977 and 1988. Separately by cohort and sex, we use a probit model to predict the probability an individual is present in the study site on January 1, 1996 using the set of baseline household and household head characteristics (which includes pre-program migration networks for the household compound), their interactions with the treatment variable, month of birth and year of birth fixed effects, and indicators for whether an individual was from a village that experienced erosion or was exposed to the Meghna Dhonnogoda Irrigation Project.

The weight to account for attrition between MHSS1 and MHSS2 is constructed in a similar manner. We estimate the probability of non-attrition between the two survey waves for each cohort-sex group using a probit model and the same set of covariates. The resulting attrition weight is the inverse of the product of the two probabilities.